

Derivations for Empirical Industrial Organization

Chapters 3–9

Following the course notes for ECON5630 Topics in Empirical Industrial Organization by Kohei Kawaguchi¹

Assisted by AI

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Contents

1	Notation and Roadmap	4
2	Chapter 3: Production and Cost Function Estimation	4
2.1	Cobb-Douglas Production: Log-Linear Form	4
2.2	Static Labor Choice and the Sign of OLS Bias	5
2.3	Selection Bias from Exit	5
2.4	Panel Differencing under Fixed Firm Heterogeneity	6
2.5	Instrumental Variables and GMM Moments	6
2.6	Olley-Pakes: Proxy Inversion	7
2.7	Olley-Pakes: Second Stage and Productivity Recovery	8
2.8	Selection Correction in OP	8
2.9	Zero Investment and Levinsohn-Petrin	9
2.10	Collinearity in OP/LP and the ACF Critique	10
2.11	Share Regression from a Static First-Order Condition	10
2.12	Cost-Function Duality	11
2.13	Translog Cost Shares and Restrictions	12
2.14	Revealed Marginal Cost: Single-Product Monopoly	12
2.15	Revealed Marginal Costs: Multi-Product Monopoly	13
2.16	Revealed Marginal Costs: Bertrand Oligopoly	13
3	Chapter 4: Demand Function Estimation	14
3.1	Utility Maximization, Expenditure Function, and Shephard's Lemma	14
3.2	Roy's Identity	14
3.3	AIDS from the PIGLOG Expenditure Function	15
3.4	AIDS Adding-Up, Homogeneity, and Symmetry	16
3.5	Discrete Choice: Budget Reduction and Choice-Specific Indirect Utility	16
3.6	Aggregation of Individual Demand	17
3.7	Horizontal Differentiation: Location Model	17
3.8	Vertical Differentiation	17
3.9	Multinomial Logit Choice Probability: Full Derivation	18
3.10	Inclusive Value under Logit	19
3.11	Logit Inversion, MLE, and NLLS	19
3.12	Logit Price Elasticities	20
3.13	Mixed Logit Choice Probability and Elasticities	20
3.14	Monte Carlo and Gauss-Hermite Integration	21
3.15	Nested Logit Choice Probability and Decomposition	21
3.16	Nested Logit Linearization	22
3.17	BLP Mean-Utility Inversion	22
3.18	BLP GMM and Concentrating Out Linear Parameters	23
3.19	Control Function for Price Endogeneity	23

4	Chapter 5: Merger Analysis	24
4.1	Conduct Parameter in a Homogeneous-Goods Market	24
4.2	Linear Demand, Linear Cost, and Nonidentification of Conduct	24
4.3	Identification with Cost Data	25
4.4	Identification with a Demand Rotator	25
4.5	Differentiated Products: Conduct Matrix	25
4.6	Pre-Merger Bertrand FOCs and Marginal Cost Recovery	26
4.7	Post-Merger Simulation	26
4.8	Consumer Surplus in Logit and Mixed Logit	26
5	Chapter 6: Entry and Exit	27
5.1	Monopoly Entry and Nonidentification	27
5.2	Identification with Known Distribution or Known Profit	27
5.3	Identification under Homogeneity of Variable Profits	27
5.4	Complete-Information Homogeneous Oligopoly Entry	28
5.5	Ordered-Response Likelihood	28
5.6	Two-Firm Complete-Information Entry Game	28
5.7	Moment Inequalities with Multiple Equilibria	29
5.8	Incomplete-Information Entry Thresholds	29
6	Chapter 7: Dynamic Decision Model	30
6.1	Bellman Equation for a Single Agent	30
6.2	Logit CCPs and Inclusive-Value Bellman Operator	30
6.3	Identification Logic	31
6.4	Nested Fixed-Point Estimation	31
6.5	CCP Approach: Main Idea	31
6.6	Expected Shock Conditional on Choice under Logit	31
6.7	CCP Value Function in Matrix Form	32
6.8	CCP Optimality Equation	33
6.9	Finite-Mixture Identification: Factorization	33
6.10	EM Algorithm for Unobserved Dynamic States	33
7	Chapter 8: Dynamic Game	34
7.1	Multi-Agent Value Function	34
7.2	Choice-Specific Value and Equilibrium CCP	34
7.3	Identification Order Condition	35
7.4	CCP Estimation for Dynamic Games	35
7.5	Continuous Controls: Inverting the Policy Function	35
7.6	BBL Inequality Estimator	36
7.7	Dynamic Oligopoly with Entry, Exit, and Investment	36
7.8	Oblivious Equilibrium	36
8	Chapter 9: Auctions	36
8.1	Second-Price/Button Auction under Symmetric Information	36

8.2	Second-Price Auction with Reserve Price	37
8.3	IPV Button Auction with Winning Bid Only	37
8.4	Optimal Reserve Price in IPV Second-Price Auction	38
8.5	First-Price IPV Auction: Equilibrium Differential Equation	38
8.6	Solving the First-Price Differential Equation	38
8.7	First-Price MLE with Winning Bids	39
8.8	GPV Nonparametric Inversion	40
8.9	Set Identification in English Auctions with Bid Increments	40
A	Supplementary Worked Derivations and Algebraic Details	41
A.1	Chapter 3 Addendum: OLS Bias Written as an Omitted-Variable Bias	41
A.2	Chapter 3 Addendum: Why OP Does Not Identify the Capital Coefficient in the First Stage	41
A.3	Chapter 3 Addendum: OP Second Stage as a Nonlinear Least-Squares Problem	42
A.4	Chapter 3 Addendum: Cost Heterogeneity Regression	42
A.5	Chapter 4 Addendum: Logit IIA and the Red-Bus/Blue-Bus Algebra	43
A.6	Chapter 4 Addendum: Logit Markups under Single-Product Bertrand	43
A.7	Chapter 4 Addendum: BLP Contraction Mapping Intuition	43
A.8	Chapter 4 Addendum: Instruments in Differentiated-Products Demand	44
A.9	Chapter 5 Addendum: Multiproduct Bertrand in Matrix Form	44
A.10	Chapter 5 Addendum: Logit Merger Matrix Elements	45
A.11	Chapter 5 Addendum: Why Demand Rotation Identifies Conduct	45
A.12	Chapter 6 Addendum: Ordered Entry Probabilities	45
A.13	Chapter 6 Addendum: The Multiple-Equilibrium Rectangle	46
A.14	Chapter 6 Addendum: Incomplete-Information Equilibrium as Fixed Point in Entry Probabilities	46
A.15	Chapter 7 Addendum: Bellman Contraction Proof Sketch	46
A.16	Chapter 7 Addendum: Hotz-Miller Inversion for Logit	47
A.17	Chapter 7 Addendum: Partial Likelihood in NFXP	47
A.18	Chapter 8 Addendum: From Individual CCPs to Joint Action Probabilities	47
A.19	Chapter 8 Addendum: Why Multiple Equilibria Complicate Likelihood	48
A.20	Chapter 8 Addendum: Forward Simulation for BBL Values	48
A.21	Chapter 9 Addendum: Why First-Price Bids Are Below Values	48
A.22	Chapter 9 Addendum: Observed Heterogeneity in Auctions	49
A.23	Chapter 9 Addendum: Winner's Curse in Common-Value Auctions	49
B	Compact Checklist of Estimation Moments	49
C	Common Normalizations	50

1 Notation and Roadmap

This document expands the main algebraic and econometric derivations in Chapters 3–9. I keep notation close to the lecture notes. Lower-case variables usually denote natural logarithms, for example $y = \log Y$, $l = \log L$, and $k = \log K$. For random variables, $\mathbb{E}[\cdot]$ denotes expectation, $\mathbb{P}\{\cdot\}$ denotes probability, and $\mathbf{1}\{\cdot\}$ denotes an indicator.

The term “identified” is used in the population sense. An object is identified if it is pinned down by the joint distribution of observables and the maintained assumptions. Estimation then replaces population objects by sample analogues, series approximations, simulations, or numerical fixed points.

The document is organized by topic. Section 2 derives production and cost-function formulas. Section 3 derives continuous and discrete demand systems, including logit, nested logit, mixed logit, and BLP inversion. Section 4 derives conduct and merger-simulation equations. Section 5 derives entry and exit likelihoods and moment inequalities. Section 6 derives single-agent dynamic decision models and the CCP approach. Section 7 derives dynamic-game CCP formulas, continuous-control logic, and oblivious equilibrium. Section 8 derives auction equilibrium and estimation formulas.

2 Chapter 3: Production and Cost Function Estimation

2.1 Cobb-Douglas Production: Log-Linear Form

Start with the Cobb-Douglas production function for firm j in period t ,

$$Y_{jt} = A_{jt} L_{jt}^{\beta_l} K_{jt}^{\beta_k}, \quad A_{jt} > 0. \quad (1)$$

Taking logarithms term by term gives

$$\log Y_{jt} = \log A_{jt} + \beta_l \log L_{jt} + \beta_k \log K_{jt}. \quad (2)$$

Let

$$y_{jt} = \log Y_{jt}, \quad l_{jt} = \log L_{jt}, \quad k_{jt} = \log K_{jt}.$$

Decompose productivity into a constant and an unobserved disturbance,

$$\log A_{jt} = \beta_0 + \epsilon_{jt}. \quad (3)$$

Substituting this decomposition into (2) yields the linear regression-looking equation

$$y_{jt} = \beta_0 + \beta_l l_{jt} + \beta_k k_{jt} + \epsilon_{jt}. \quad (4)$$

The linearity is useful, but it does not by itself justify OLS. The economic problem is that l_{jt} and sometimes k_{jt} are choices, while ϵ_{jt} contains information observed by the firm but not by the econometrician.

2.2 Static Labor Choice and the Sign of OLS Bias

Suppose capital K_{jt} is predetermined and the firm chooses labor after observing ϵ_{jt} . With output price p_{jt} and wage w_{jt} , current profit is

$$\Pi(L) = p_{jt} \exp(\beta_0 + \epsilon_{jt}) L^{\beta_l} K_{jt}^{\beta_k} - w_{jt} L. \quad (5)$$

The first-order condition for an interior optimum is

$$\begin{aligned} \frac{\partial \Pi(L)}{\partial L} &= p_{jt} \exp(\beta_0 + \epsilon_{jt}) \beta_l L^{\beta_l - 1} K_{jt}^{\beta_k} - w_{jt} = 0, \\ L^{\beta_l - 1} &= \frac{w_{jt}}{p_{jt} \beta_l \exp(\beta_0 + \epsilon_{jt}) K_{jt}^{\beta_k}}. \end{aligned} \quad (6)$$

Because $\beta_l - 1 = -(1 - \beta_l)$, rewrite the last expression as

$$L^{1 - \beta_l} = \frac{p_{jt} \beta_l \exp(\beta_0 + \epsilon_{jt}) K_{jt}^{\beta_k}}{w_{jt}}. \quad (7)$$

Therefore

$$L_{jt} = \left[\frac{p_{jt}}{w_{jt}} \beta_l \exp(\beta_0 + \epsilon_{jt}) K_{jt}^{\beta_k} \right]^{1/(1 - \beta_l)}. \quad (8)$$

If $0 < \beta_l < 1$, L_{jt} is increasing in ϵ_{jt} . Taking logs,

$$l_{jt} = \frac{1}{1 - \beta_l} (\log p_{jt} - \log w_{jt} + \log \beta_l + \beta_0 + \epsilon_{jt} + \beta_k k_{jt}). \quad (9)$$

Thus, after controlling for prices and capital, l_{jt} is positively related to ϵ_{jt} . In the population least-squares formula for (4),

$$\text{plim } \hat{\beta}_{OLS} = \beta + \left(\mathbb{E}[X_{jt} X'_{jt}] \right)^{-1} \mathbb{E}[X_{jt} \epsilon_{jt}], \quad X_{jt} = (1, l_{jt}, k_{jt})', \quad (10)$$

so if $\mathbb{E}[l_{jt} \epsilon_{jt}] > 0$, the labor coefficient is biased upward. Intuitively, high-productivity firms choose more labor and also produce more output. OLS cannot tell which part of the output increase is caused by labor and which part is caused by productivity.

2.3 Selection Bias from Exit

Let ω_{jt} be the productivity component observed before the exit decision. Suppose survival is described by a threshold rule

$$\chi_{jt} = \mathbf{1}\{\omega_{jt} \geq \omega^*(k_{jt})\}, \quad (11)$$

where $\chi_{jt} = 1$ means the firm stays in the sample. If larger capital makes survival easier, then $\omega^*(k)$ is decreasing in k . The conditional productivity among observed firms is

$$\mathbb{E}[\omega_{jt} \mid k_{jt}, \chi_{jt} = 1] = \mathbb{E}[\omega_{jt} \mid \omega_{jt} \geq \omega^*(k_{jt})]. \quad (12)$$

A higher k lowers the truncation point. Hence observed high-capital firms may include lower-productivity firms that would have exited if capital were lower. This induces a negative association between k_{jt} and ω_{jt} conditional on survival. The direction is different from the simultaneity bias above, and it can bias the capital coefficient downward.

2.4 Panel Differencing under Fixed Firm Heterogeneity

Assume the production error has a firm fixed component and a remaining shock,

$$\epsilon_{jt} = \mu_j + \eta_{jt}, \quad \mathbb{E}[\eta_{jt} \mid l_{js}, k_{js}, s \leq t] = 0. \quad (13)$$

Then

$$y_{jt} = \beta_0 + \beta_l l_{jt} + \beta_k k_{jt} + \mu_j + \eta_{jt}. \quad (14)$$

Subtracting the period $t - 1$ equation from the period t equation gives

$$\Delta y_{jt} = \beta_l \Delta l_{jt} + \beta_k \Delta k_{jt} + \Delta \eta_{jt}, \quad (15)$$

where $\Delta y_{jt} = y_{jt} - y_{j,t-1}$ and similarly for inputs. The fixed effect and intercept disappear. If

$$\mathbb{E}[\Delta \eta_{jt} \mid \Delta l_{jt}, \Delta k_{jt}] = 0, \quad (16)$$

then β_l and β_k are identified from the population moment

$$\mathbb{E} \left[\begin{pmatrix} \Delta l_{jt} \\ \Delta k_{jt} \end{pmatrix} (\Delta y_{jt} - \beta_l \Delta l_{jt} - \beta_k \Delta k_{jt}) \right] = 0. \quad (17)$$

The important caveat is measurement error. If $l_{jt}^{obs} = l_{jt} + u_{jt}$ with serially independent error u_{jt} , then

$$\Delta l_{jt}^{obs} = \Delta l_{jt} + u_{jt} - u_{j,t-1}. \quad (18)$$

The measurement-error variance in the differenced input is $2 \text{Var}(u_{jt})$. Differencing removes fixed heterogeneity but can magnify attenuation bias.

2.5 Instrumental Variables and GMM Moments

Let the production equation be

$$y = X\beta + \epsilon, \quad (19)$$

where X contains endogenous inputs. Suppose instruments Z satisfy

$$\mathbb{E}[Z'\epsilon] = 0, \quad \text{rank } \mathbb{E}[Z'X] = \dim(\beta). \quad (20)$$

Then the identifying moment is

$$\mathbb{E}[Z'(y - X\beta)] = 0. \quad (21)$$

If the model is exactly identified, the population solution is

$$\beta = (\mathbb{E}[Z'X])^{-1} \mathbb{E}[Z'y]. \quad (22)$$

If there are more instruments than parameters, estimate by GMM:

$$\hat{\beta} = \arg \min_b \hat{g}(b)' W \hat{g}(b), \quad \hat{g}(b) = \frac{1}{n} \sum_{i=1}^n Z_i(y_i - X_i'b). \quad (23)$$

Input prices are natural instruments when they shift input choices but are exogenous to firm-specific productivity. Lagged inputs can be instruments only under conditions that break serial correlation between current productivity and lagged decisions.

2.6 Olley-Pakes: Proxy Inversion

Decompose the production disturbance as

$$y_{jt} = \beta_0 + \beta_l l_{jt} + \beta_k k_{jt} + \omega_{jt} + \eta_{jt}, \quad (24)$$

where ω_{jt} is an anticipated productivity shock known to the firm when it chooses inputs, and η_{jt} is an ex-post shock not known when inputs are chosen.

Key assumptions. The OP proxy argument requires: (i) a scalar unobservable ω_{jt} ; (ii) investment follows a policy $i_{jt} = f_t(k_{jt}, \omega_{jt})$; (iii) f_t is strictly increasing in ω_{jt} for each k_{jt} ; and (iv) the transitory shock satisfies $\mathbb{E}[\eta_{jt} | I_{jt}] = 0$.

Strict monotonicity permits inversion:

$$\omega_{jt} = h_t(k_{jt}, i_{jt}). \quad (25)$$

Substitute (25) into (24):

$$\begin{aligned} y_{jt} &= \beta_l l_{jt} + \beta_0 + \beta_k k_{jt} + h_t(k_{jt}, i_{jt}) + \eta_{jt} \\ &= \beta_l l_{jt} + \phi_t(k_{jt}, i_{jt}) + \eta_{jt}, \end{aligned} \quad (26)$$

where

$$\phi_t(k, i) = \beta_0 + \beta_k k + h_t(k, i). \quad (27)$$

The first-stage population moment is

$$\mathbb{E} [y_{jt} - \beta_l l_{jt} - \phi_t(k_{jt}, i_{jt}) | l_{jt}, k_{jt}, i_{jt}] = 0. \quad (28)$$

In practice, approximate ϕ_t by basis functions q_p :

$$\phi_t(k, i) \approx \sum_{p=1}^P \gamma_{pt} q_p(k, i). \quad (29)$$

For example, a second-order polynomial is

$$\phi_t(k, i) \approx \gamma_0 + \gamma_1 k + \gamma_2 i + \gamma_3 k^2 + \gamma_4 i^2 + \gamma_5 ki. \quad (30)$$

2.7 Olley-Pakes: Second Stage and Productivity Recovery

Assume anticipated productivity follows a first-order Markov process,

$$\omega_{jt} = g(\omega_{j,t-1}) + v_{jt}, \quad \mathbb{E}[v_{jt} \mid I_{j,t-1}] = 0. \quad (31)$$

From the first-stage control,

$$\omega_{j,t-1} = \phi_{t-1}(k_{j,t-1}, i_{j,t-1}) - \beta_0 - \beta_k k_{j,t-1}. \quad (32)$$

Subtract $\beta_l l_{jt}$ from the production equation:

$$\begin{aligned} y_{jt} - \beta_l l_{jt} &= \beta_0 + \beta_k k_{jt} + \omega_{jt} + \eta_{jt} \\ &= \beta_0 + \beta_k k_{jt} + g(\omega_{j,t-1}) + v_{jt} + \eta_{jt} \\ &= \beta_0 + \beta_k k_{jt} + g(\phi_{t-1}(k_{j,t-1}, i_{j,t-1}) - \beta_0 - \beta_k k_{j,t-1}) + v_{jt} + \eta_{jt}. \end{aligned} \quad (33)$$

Under the timing assumptions, current capital and lagged investment/capital are valid conditioning variables, so

$$\mathbb{E}[v_{jt} + \eta_{jt} \mid k_{jt}, k_{j,t-1}, i_{j,t-1}] = 0. \quad (34)$$

The second stage identifies β_0 , β_k , and the nonparametric function $g(\cdot)$.

Once $\hat{\phi}_t$, $\hat{\beta}_0$, and $\hat{\beta}_k$ are available, recover anticipated productivity by

$$\hat{\omega}_{jt} = \hat{\phi}_t(k_{jt}, i_{jt}) - \hat{\beta}_0 - \hat{\beta}_k k_{jt}. \quad (35)$$

This follows directly from the definition $\phi_t = \beta_0 + \beta_k k + \omega$.

2.8 Selection Correction in OP

Survival is described by

$$\chi_{jt} = \mathbf{1}\{\omega_{jt} \geq \omega^*(k_{jt})\}. \quad (36)$$

The first stage needs no correction if the transitory shock is realized after the survival decision:

$$\begin{aligned} \mathbb{E}[y_{jt} \mid l_{jt}, k_{jt}, i_{jt}, \chi_{jt} = 1] &= \beta_l l_{jt} + \phi_t(k_{jt}, i_{jt}) + \mathbb{E}[\eta_{jt} \mid l_{jt}, k_{jt}, i_{jt}, \chi_{jt} = 1] \\ &= \beta_l l_{jt} + \phi_t(k_{jt}, i_{jt}). \end{aligned} \quad (37)$$

The second stage must be corrected because survival truncates ω_{jt} and thus affects the innovation v_{jt} . Conditional on lagged productivity,

$$\begin{aligned}\mathbb{E}[\omega_{jt} \mid \omega_{j,t-1}, \chi_{jt} = 1] &= \mathbb{E}[\omega_{jt} \mid \omega_{j,t-1}, \omega_{jt} \geq \omega^*(k_{jt})] \\ &\equiv \tilde{g}(\omega_{j,t-1}, \omega^*(k_{jt})).\end{aligned}\quad (38)$$

Define the continuation probability

$$\begin{aligned}P_{jt} &= \mathbb{P}(\chi_{jt} = 1 \mid I_{j,t-1}) \\ &= \mathbb{P}\{g(\omega_{j,t-1}) + v_{jt} \geq \omega^*(k_{jt}) \mid I_{j,t-1}\}.\end{aligned}\quad (39)$$

Since $k_{jt} = \kappa(k_{j,t-1}, i_{j,t-1})$, P_{jt} is a function of lagged capital and investment. If the mapping from the threshold $\omega^*(k_{jt})$ to P_{jt} is invertible conditional on $\omega_{j,t-1}$, then

$$\omega^*(k_{jt}) = \psi(P_{jt}, \omega_{j,t-1}).\quad (40)$$

Substitute this into (38):

$$\mathbb{E}[\omega_{jt} \mid \omega_{j,t-1}, \chi_{jt} = 1] = \tilde{g}(\omega_{j,t-1}, P_{jt}).\quad (41)$$

The corrected second-stage equation is

$$y_{jt} - \beta_l l_{jt} = \beta_0 + \beta_k k_{jt} + \tilde{g}(\phi_{t-1}(k_{j,t-1}, i_{j,t-1}) - \beta_0 - \beta_k k_{j,t-1}, P_{jt}) + \text{mean-zero error}.\quad (42)$$

Hence the OP selection correction adds the estimated propensity score as an argument of the nonparametric control.

2.9 Zero Investment and Levinsohn-Petrin

If many observations have $i_{jt} = 0$, the investment policy is not globally invertible: multiple productivity values may map to zero investment. One solution is to drop zero-investment observations, but this loses information. Levinsohn-Petrin uses an intermediate input, such as materials, as the proxy.

Let gross output be

$$y_{jt} = \beta_0 + \beta_l l_{jt} + \beta_k k_{jt} + \beta_m m_{jt} + \omega_{jt} + \eta_{jt}.\quad (43)$$

Assume materials are chosen according to a strictly increasing policy

$$m_{jt} = m(k_{jt}, \omega_{jt}), \quad \omega_{jt} = h(m_{jt}, k_{jt}).\quad (44)$$

The first stage is

$$y_{jt} = \beta_l l_{jt} + \underbrace{\beta_0 + \beta_k k_{jt} + \beta_m m_{jt} + h(m_{jt}, k_{jt})}_{\phi(m_{jt}, k_{jt})} + \eta_{jt},$$

$$= \beta_l l_{jt} + \phi(m_{jt}, k_{jt}) + \eta_{jt}. \quad (45)$$

Thus

$$\mathbb{E}[y_{jt} - \beta_l l_{jt} - \phi(m_{jt}, k_{jt}) \mid l_{jt}, m_{jt}, k_{jt}] = 0. \quad (46)$$

The second stage uses

$$\omega_{j,t-1} = \phi(m_{j,t-1}, k_{j,t-1}) - \beta_0 - \beta_k k_{j,t-1} - \beta_m m_{j,t-1}. \quad (47)$$

Therefore

$$\begin{aligned} y_{jt} - \beta_l l_{jt} &= \beta_0 + \beta_k k_{jt} + \beta_m m_{jt} \\ &+ g(\phi(m_{j,t-1}, k_{j,t-1}) - \beta_0 - \beta_k k_{j,t-1} - \beta_m m_{j,t-1}) + v_{jt} + \eta_{jt}. \end{aligned} \quad (48)$$

Because current materials are chosen after observing current productivity innovations, a valid moment should not condition on m_{jt} . A common moment is

$$\mathbb{E}[v_{jt} + \eta_{jt} \mid k_{jt}, m_{j,t-1}, k_{j,t-1}] = 0. \quad (49)$$

2.10 Collinearity in OP/LP and the ACF Critique

If labor is itself a deterministic policy of the same state variables used to proxy productivity, then it may not have independent variation in the first stage. Suppose

$$l_{jt} = \ell(k_{jt}, \omega_{jt}), \quad \omega_{jt} = h(i_{jt}, k_{jt}). \quad (50)$$

Then

$$l_{jt} = \ell(k_{jt}, h(i_{jt}, k_{jt})) \equiv \tilde{\ell}(i_{jt}, k_{jt}). \quad (51)$$

The first stage becomes

$$\begin{aligned} y_{jt} &= \beta_l \tilde{\ell}(i_{jt}, k_{jt}) + \phi(i_{jt}, k_{jt}) + \eta_{jt} \\ &= \tilde{\phi}(i_{jt}, k_{jt}) + \eta_{jt}. \end{aligned} \quad (52)$$

Because the unknown nonparametric function can absorb $\beta_l \tilde{\ell}(i, k)$, β_l is not separately identified in that stage. Akerberg-Caves-Frazer style timing assumptions treat labor as chosen after an additional shock or move labor into the proxy function, then identify all coefficients in a second stage using lagged instruments.

2.11 Share Regression from a Static First-Order Condition

Let physical output be

$$Y = F(L, K) \exp(\omega) \exp(\eta), \quad (53)$$

where ω is known when labor is chosen and η is not. With competitive output and labor markets, the firm chooses labor by maximizing expected current profit. The FOC is

$$PF_L(L, K) \exp(\omega) \mathbb{E}[\exp(\eta)] = w. \quad (54)$$

Multiply both sides by L and divide by realized revenue $PY = PF(L, K) \exp(\omega) \exp(\eta)$:

$$\frac{F_L(L, K) L \mathbb{E}[\exp(\eta)]}{F(L, K) \exp(\eta)} = \frac{wL}{PY}. \quad (55)$$

Taking logs gives the labor cost-share equation

$$s \equiv \log \frac{wL}{PY} = \log \left(\frac{F_L(L, K) L}{F(L, K)} \right) + \log \mathbb{E}[\exp(\eta)] - \eta. \quad (56)$$

For Cobb-Douglas $F(L, K) = L^{\beta_l} K^{\beta_k}$,

$$F_L(L, K) = \beta_l L^{\beta_l-1} K^{\beta_k}, \quad \frac{F_L(L, K) L}{F(L, K)} = \beta_l. \quad (57)$$

Thus

$$s = \log \beta_l + \log \mathbb{E}[\exp(\eta)] - \eta. \quad (58)$$

The anticipated productivity term ω cancels. This is why a cost-share equation can identify static elasticities that are hard to identify from the production equation alone.

2.12 Cost-Function Duality

Given a production function $F(x)$ and input price vector p , the cost function is

$$C(y, p) = \min_x p'x \quad \text{s.t.} \quad F(x) \geq y. \quad (59)$$

The Lagrangian is

$$\mathcal{L}(x, \lambda) = p'x + \lambda[y - F(x)]. \quad (60)$$

Interior first-order conditions are

$$p_i = \lambda F_i(x), \quad i = 1, \dots, J. \quad (61)$$

The resulting cost function has several restrictions:

- Homogeneity of degree one in input prices:

$$C(y, \alpha p) = \min_x \alpha p'x = \alpha C(y, p), \quad \alpha > 0. \quad (62)$$

- Monotonicity in output: higher y shrinks the feasible set, so minimized cost cannot fall.
- Concavity in input prices: $C(y, p) = \inf_{x \in X(y)} p'x$ is the pointwise infimum of linear functions of p .

Shephard's lemma gives compensated input demand:

$$x_i(y, p) = \frac{\partial C(y, p)}{\partial p_i}. \quad (63)$$

2.13 Translog Cost Shares and Restrictions

A translog cost function is

$$\log C = \alpha_0 + \alpha'_p \log p + \alpha_y \log y + \frac{1}{2}(\log p)' B_{pp} \log p + (\log p)' \beta_{py} \log y + \frac{1}{2} \beta_{yy} (\log y)^2. \quad (64)$$

By Shephard's lemma, the cost share of input i is

$$s_i = \frac{p_i x_i}{C} = \frac{\partial \log C}{\partial \log p_i}. \quad (65)$$

Differentiate (64):

$$s_i = \alpha_i + \sum_j B_{ij} \log p_j + \beta_{iy} \log y. \quad (66)$$

For homogeneity of degree one, $\sum_i s_i = 1$ for all (p, y) . Hence

$$\sum_i \alpha_i = 1, \quad \sum_i B_{ij} = 0 \text{ for every } j, \quad \sum_i \beta_{iy} = 0. \quad (67)$$

Symmetry of the Hessian requires $B_{ij} = B_{ji}$. Concavity in input prices imposes curvature restrictions on the share-elasticity matrix.

2.14 Revealed Marginal Cost: Single-Product Monopoly

A monopolist faces demand $D(p)$ and cost $C(q)$. Profit is

$$\Pi(p) = pD(p) - C(D(p)). \quad (68)$$

Differentiate:

$$\frac{d\Pi(p)}{dp} = D(p) + pD'(p) - C'(D(p))D'(p). \quad (69)$$

Set the derivative equal to zero and denote marginal cost by $MC(q) = C'(q)$:

$$D(p) + pD'(p) - MC(D(p))D'(p) = 0. \quad (70)$$

Solving for marginal cost gives

$$MC(D(p)) = p + \frac{D(p)}{D'(p)}. \quad (71)$$

Because $D'(p) < 0$, the markup is $p - MC(D(p)) = -D(p)/D'(p) > 0$.

2.15 *Revealed Marginal Costs: Multi-Product Monopoly*

Let $D(p) = (D_1(p), \dots, D_J(p))'$ and $C(q)$ be a joint cost function. Profit is

$$\Pi(p) = \sum_{j=1}^J p_j D_j(p) - C(D(p)). \quad (72)$$

The FOC with respect to p_i is

$$D_i(p) + \sum_{j=1}^J p_j \frac{\partial D_j(p)}{\partial p_i} - \sum_{j=1}^J MC_j(D(p)) \frac{\partial D_j(p)}{\partial p_i} = 0. \quad (73)$$

Define the Jacobian matrix $\Delta_D(p)$ by

$$[\Delta_D(p)]_{ij} = \frac{\partial D_j(p)}{\partial p_i}. \quad (74)$$

Stacking FOCs gives

$$D(p) + \Delta_D(p)p = \Delta_D(p)mc. \quad (75)$$

If $\Delta_D(p)$ is invertible,

$$mc = p + \Delta_D(p)^{-1}D(p). \quad (76)$$

2.16 *Revealed Marginal Costs: Bertrand Oligopoly*

For a single-product firm j that chooses p_j taking p_{-j} as given,

$$\Pi_j(p) = (p_j - mc_j)D_j(p). \quad (77)$$

The FOC is

$$D_j(p) + (p_j - mc_j) \frac{\partial D_j(p)}{\partial p_j} = 0, \quad (78)$$

so

$$mc_j = p_j + \frac{D_j(p)}{\partial D_j(p) / \partial p_j}. \quad (79)$$

For multiproduct firms, the FOC for product k owned by firm f is

$$D_k(p) + \sum_{j \in \mathcal{J}_f} (p_j - mc_j) \frac{\partial D_j(p)}{\partial p_k} = 0. \quad (80)$$

This is the basis of the merger-simulation formulas in Chapter 5.

3 Chapter 4: Demand Function Estimation

3.1 Utility Maximization, Expenditure Function, and Shephard's Lemma

For continuous goods, the Marshallian problem is

$$v(p, x) = \max_q u(q) \quad \text{s.t.} \quad p'q \leq x. \quad (81)$$

The dual expenditure function is

$$e(u, p) = \min_q p'q \quad \text{s.t.} \quad u(q) \geq u. \quad (82)$$

Let $h_i(u, p)$ be Hicksian demand. The envelope theorem gives Shephard's lemma:

$$\frac{\partial e(u, p)}{\partial p_i} = h_i(u, p). \quad (83)$$

If $x = e(u, p)$, the indirect utility is the inverse in the utility argument,

$$u = v(p, x) = e^{-1}(p, x). \quad (84)$$

Marshallian demand is Hicksian demand evaluated at indirect utility:

$$d_i(p, x) = h_i(v(p, x), p). \quad (85)$$

The expenditure share equation follows from Shephard's lemma:

$$\frac{\partial \log e(u, p)}{\partial \log p_i} = \frac{p_i}{e(u, p)} \frac{\partial e(u, p)}{\partial p_i} = \frac{p_i h_i(u, p)}{x} \equiv w_i. \quad (86)$$

3.2 Roy's Identity

The indirect utility function is

$$v(p, x) = \max_q u(q) \quad \text{s.t.} \quad p'q \leq x. \quad (87)$$

Let $q^*(p, x)$ be the optimum and λ the multiplier on the budget. The Lagrangian is

$$\mathcal{L}(q, \lambda) = u(q) + \lambda(x - p'q). \quad (88)$$

The envelope theorem gives

$$\frac{\partial v(p, x)}{\partial x} = \lambda, \quad \frac{\partial v(p, x)}{\partial p_i} = -\lambda q_i^*(p, x). \quad (89)$$

Divide the second expression by the first:

$$q_i^*(p, x) = -\frac{\partial v(p, x)/\partial p_i}{\partial v(p, x)/\partial x}. \quad (90)$$

This is Roy's identity.

3.3 AIDS from the PIGLOG Expenditure Function

The PIGLOG class is

$$\log e(u, p) = (1 - u) \log a(p) + u \log b(p). \quad (91)$$

AIDS specifies

$$\log a(p) = \alpha_0 + \sum_k \alpha_k \log p_k + \frac{1}{2} \sum_k \sum_j \gamma_{kj}^* \log p_k \log p_j, \quad (92)$$

$$\log b(p) = \log a(p) + \beta_0 \prod_k p_k^{\beta_k}. \quad (93)$$

Differentiate $\log e$ with respect to $\log p_i$:

$$\begin{aligned} w_i &= \frac{\partial \log e(u, p)}{\partial \log p_i} \\ &= (1 - u) \frac{\partial \log a(p)}{\partial \log p_i} + u \frac{\partial \log b(p)}{\partial \log p_i} \\ &= \frac{\partial \log a(p)}{\partial \log p_i} + u \frac{\partial \{\log b(p) - \log a(p)\}}{\partial \log p_i}. \end{aligned} \quad (94)$$

From (92),

$$\frac{\partial \log a(p)}{\partial \log p_i} = \alpha_i + \frac{1}{2} \sum_j \gamma_{ij}^* \log p_j + \frac{1}{2} \sum_k \gamma_{ki}^* \log p_k = \alpha_i + \sum_j \gamma_{ij} \log p_j, \quad (95)$$

where $\gamma_{ij} = (\gamma_{ij}^* + \gamma_{ji}^*)/2$. From (93),

$$\frac{\partial}{\partial \log p_i} \left[\beta_0 \prod_k p_k^{\beta_k} \right] = \beta_i \beta_0 \prod_k p_k^{\beta_k}. \quad (96)$$

Hence

$$w_i = \alpha_i + \sum_j \gamma_{ij} \log p_j + \beta_i u \beta_0 \prod_k p_k^{\beta_k}. \quad (97)$$

To obtain the empirical share equation, use $x = e(u, p)$ and solve the PIGLOG expression for utility. From

$$\log x = \log a(p) + u [\log b(p) - \log a(p)] = \log a(p) + u\beta_0 \prod_k p_k^{\beta_k}, \quad (98)$$

we get

$$u\beta_0 \prod_k p_k^{\beta_k} = \log x - \log a(p). \quad (99)$$

Let $\log P \equiv \log a(p)$. Substituting into (97) yields

$$w_i = \alpha_i + \sum_j \gamma_{ij} \log p_j + \beta_i (\log x - \log P) = \alpha_i + \sum_j \gamma_{ij} \log p_j + \beta_i \log \frac{x}{P}. \quad (100)$$

3.4 AIDS Adding-Up, Homogeneity, and Symmetry

Since shares sum to one, $\sum_i w_i = 1$ for all p and x . Therefore

$$\sum_i \alpha_i = 1, \quad \sum_i \gamma_{ij} = 0 \text{ for all } j, \quad \sum_i \beta_i = 0. \quad (101)$$

Homogeneity of demand of degree zero in prices and income requires

$$\sum_j \gamma_{ij} = 0 \quad \text{for every } i. \quad (102)$$

Slutsky symmetry requires

$$\gamma_{ij} = \gamma_{ji}. \quad (103)$$

These restrictions can be imposed or tested in the share-equation system.

3.5 Discrete Choice: Budget Reduction and Choice-Specific Indirect Utility

Let alternative 0 be the outside good, normalize $p_0 = 1$, and suppose the consumer can choose at most one inside good. If the consumer chooses inside good $j > 0$, then

$$q_0 + p_j = y_i, \quad q_j = 1, \quad q_k = 0 \text{ for } k \neq j. \quad (104)$$

Thus $q_0 = y_i - p_j$. If the consumer chooses the outside good, $q_0 = y_i$ and $q_j = 0$ for all inside goods. The original utility maximization over bundles becomes a finite maximization:

$$V(p, y_i, z_i) = \max_{j=0,1,\dots,J} v_j(p_j, y_i, z_i), \quad (105)$$

where

$$v_j(p_j, y_i, z_i) = \begin{cases} u(y_i - p_j, 0, \dots, 1, \dots, 0, z_i), & j > 0, \\ u(y_i, 0, \dots, 0, z_i), & j = 0. \end{cases} \quad (106)$$

When products are represented by characteristics x_j , write

$$v_j = v(p_j, x_j, y_i, z_i). \quad (107)$$

A common no-income-effect specification is

$$v_{ij} = \beta'_i x_j - \alpha_i p_j, \quad (108)$$

because any additive term depending only on y_i cancels across alternatives.

3.6 Aggregation of Individual Demand

Let the deterministic individual choice be

$$q_{ij}(p, x, y_i, z_i) = \mathbf{1} \left\{ j = \arg \max_{k=0, \dots, J} v(p_k, x_k, y_i, z_i) \right\}. \quad (109)$$

If (y, z) has density $f(y, z)$ in a population of size N , aggregate demand for product j is

$$\sigma_j(p, x) = N \int q_{ij}(p, x, y, z) f(y, z) dy dz. \quad (110)$$

With random utility shocks, q_{ij} is replaced by the choice probability conditional on observables.

3.7 Horizontal Differentiation: Location Model

There are two stores at locations $x_1 < x_2$ on $[0, 1]$. Consumer i at location z_i has utility

$$v_{ij} = s - t|z_i - x_j| - p_j. \quad (111)$$

Assuming the marginal consumer lies between the stores, consumer i chooses store 1 iff

$$\begin{aligned} s - t(z_i - x_1) - p_1 &\geq s - t(x_2 - z_i) - p_2, \\ -tz_i + tx_1 - p_1 &\geq -tx_2 + tz_i - p_2, \\ 2tz_i &\leq p_2 - p_1 + t(x_1 + x_2), \\ z_i &\leq \frac{p_2 - p_1}{2t} + \frac{x_1 + x_2}{2} \equiv z_1(p_1, p_2). \end{aligned} \quad (112)$$

If $z_i \sim U[0, 1]$ and all consumers buy one of the two products,

$$\sigma_1(p, x) = Nz_1(p_1, p_2), \quad \sigma_2(p, x) = N[1 - z_1(p_1, p_2)]. \quad (113)$$

3.8 Vertical Differentiation

Let product quality satisfy $x_1 < \dots < x_J$ and utility be

$$v_{ij} = z_i x_j - p_j. \quad (114)$$

Consumer i prefers product $j + 1$ to j iff

$$\begin{aligned} z_i x_{j+1} - p_{j+1} &\geq z_i x_j - p_j, \\ z_i &\geq \frac{p_{j+1} - p_j}{x_{j+1} - x_j} \equiv \Delta_j. \end{aligned} \quad (115)$$

The outside-good cutoff for buying at least the lowest relevant product is

$$\Delta_0 = \min_j \frac{p_j}{x_j}. \quad (116)$$

If products are ordered so that product j is chosen for $z_i \in [\Delta_{j-1}, \Delta_j)$, aggregate demand is

$$\sigma_j = N [F(\Delta_j) - F(\Delta_{j-1})]. \quad (117)$$

3.9 Multinomial Logit Choice Probability: Full Derivation

Let random utility be

$$U_{ij} = \delta_{ij} + \epsilon_{ij}, \quad (118)$$

where ϵ_{ij} are i.i.d. type-I extreme value with CDF $G(e) = \exp[-\exp(-e)]$ and density $g(e) = \exp[-e] \exp[-\exp(-e)]$. Product j is chosen if

$$\delta_{ij} + \epsilon_{ij} \geq \delta_{ik} + \epsilon_{ik} \quad \forall k \neq j. \quad (119)$$

Condition on $\epsilon_{ij} = e$. Then

$$\mathbb{P}(j \text{ chosen} \mid \epsilon_{ij} = e) = \prod_{k \neq j} G(e + \delta_{ij} - \delta_{ik}). \quad (120)$$

Integrate over e :

$$\begin{aligned} s_{ij} &= \int_{-\infty}^{\infty} \prod_{k \neq j} \exp\{-\exp[-(e + \delta_{ij} - \delta_{ik})]\} \exp[-e] \exp[-\exp(-e)] de \\ &= \int_{-\infty}^{\infty} \exp\left\{-\sum_{k \neq j} \exp(-e - \delta_{ij} + \delta_{ik}) - \exp(-e)\right\} \exp[-e] de \\ &= \int_{-\infty}^{\infty} \exp\left\{-\exp(-e) \sum_{k=0}^J \exp(\delta_{ik} - \delta_{ij})\right\} \exp[-e] de. \end{aligned} \quad (121)$$

Let $A_j = \sum_{k=0}^J \exp(\delta_{ik} - \delta_{ij}) = \exp(-\delta_{ij}) \sum_k \exp(\delta_{ik})$. Use the change of variables $u = \exp(-e)$, so $du = -\exp(-e) de = -u de$. As $e \rightarrow -\infty$, $u \rightarrow \infty$; as $e \rightarrow \infty$, $u \rightarrow 0$. Then

$$s_{ij} = \int_{\infty}^0 \exp(-A_j u) (-du) = \int_0^{\infty} \exp(-A_j u) du = \frac{1}{A_j}$$

$$= \frac{\exp(\delta_{ij})}{\sum_{k=0}^J \exp(\delta_{ik})}. \quad (122)$$

This is the multinomial logit formula.

3.10 Inclusive Value under Logit

For i.i.d. type-I extreme value shocks,

$$\mathbb{E} \left[\max_j \{\delta_j + \epsilon_j\} \right] = \log \left(\sum_j \exp(\delta_j) \right) + \gamma, \quad (123)$$

where γ is Euler's constant. The term

$$IV = \log \left(\sum_j \exp(\delta_j) \right) \quad (124)$$

is the inclusive value. A quick derivation uses

$$\begin{aligned} \mathbb{P} \left(\max_j \{\delta_j + \epsilon_j\} \leq m \right) &= \prod_j G(m - \delta_j) \\ &= \exp \left[- \sum_j \exp(-(m - \delta_j)) \right] \\ &= \exp \left[- \exp\{-(m - IV)\} \right], \end{aligned} \quad (125)$$

so the maximum is distributed as IV plus a standard type-I extreme value shock with mean γ .

3.11 Logit Inversion, MLE, and NLLS

With outside option normalized to $\delta_0 = 0$,

$$s_j = \frac{\exp(\delta_j)}{1 + \sum_{k=1}^J \exp(\delta_k)}, \quad s_0 = \frac{1}{1 + \sum_{k=1}^J \exp(\delta_k)}. \quad (126)$$

Therefore

$$\frac{s_j}{s_0} = \exp(\delta_j), \quad \log s_j - \log s_0 = \delta_j. \quad (127)$$

If $\delta_j = x_j' \beta - \alpha p_j$, then the market-share regression is

$$\log s_j - \log s_0 = x_j' \beta - \alpha p_j + \xi_j, \quad (128)$$

where ξ_j is an unobserved product characteristic if included. Price endogeneity arises when p_j is correlated with ξ_j .

For individual-level data, the conditional log-likelihood is

$$\ell(\theta) = \sum_i \sum_{j=0}^J q_{ij} \log s_{ij}(\theta), \quad (129)$$

where $q_{ij} = 1$ if consumer i chose product j .

3.12 Logit Price Elasticities

Let $\delta_j = \beta'x_j - \alpha p_j$. The logit share is

$$s_j = \frac{\exp(\delta_j)}{\sum_l \exp(\delta_l)}. \quad (130)$$

Differentiate with respect to p_k :

$$\frac{\partial s_j}{\partial p_k} = \frac{\partial s_j}{\partial \delta_k} \frac{\partial \delta_k}{\partial p_k}. \quad (131)$$

For logit,

$$\frac{\partial s_j}{\partial \delta_k} = s_j(\mathbf{1}\{j = k\} - s_k), \quad \frac{\partial \delta_k}{\partial p_k} = -\alpha. \quad (132)$$

Thus

$$\frac{\partial s_j}{\partial p_k} = \begin{cases} -\alpha s_j(1 - s_j), & j = k, \\ \alpha s_j s_k, & j \neq k. \end{cases} \quad (133)$$

The elasticity of share j with respect to price p_k is

$$\varepsilon_{jk} = \frac{\partial s_j}{\partial p_k} \frac{p_k}{s_j} = \begin{cases} -\alpha p_j(1 - s_j), & j = k, \\ \alpha p_k s_k, & j \neq k. \end{cases} \quad (134)$$

The cross-elasticity depends only on the other product's share, producing the IIA substitution pattern.

3.13 Mixed Logit Choice Probability and Elasticities

Let individual coefficients be random:

$$\delta_{ij} = \beta'_i x_j - \alpha_i p_j, \quad (\beta_i, \alpha_i) \sim f(\beta, \alpha \mid y_i, z_i). \quad (135)$$

Conditional on (β_i, α_i) , the choice probability is logit:

$$s_{ij}(\beta_i, \alpha_i) = \frac{\exp(\beta'_i x_j - \alpha_i p_j)}{\sum_k \exp(\beta'_i x_k - \alpha_i p_k)}. \quad (136)$$

The unconditional probability integrates over unobserved types:

$$s_j(p, x, y_i, z_i) = \int s_{ij}(\beta, \alpha) f(\beta, \alpha \mid y_i, z_i) d\beta d\alpha. \quad (137)$$

The own and cross-price derivatives are

$$\frac{\partial s_j}{\partial p_k} = \begin{cases} - \int \alpha s_{ij}(1 - s_{ij}) f(\beta, \alpha) d\beta d\alpha, & j = k, \\ \int \alpha s_{ij} s_{ik} f(\beta, \alpha) d\beta d\alpha, & j \neq k. \end{cases} \quad (138)$$

Thus

$$\varepsilon_{jk} = \frac{p_k}{s_j} \frac{\partial s_j}{\partial p_k}. \quad (139)$$

Unlike simple logit, these elasticities depend on the full distribution of consumer heterogeneity.

3.14 Monte Carlo and Gauss-Hermite Integration

If no closed form exists for

$$\int f(v) \varphi(v) dv, \quad (140)$$

simulated integration draws $v_r, r = 1, \dots, R$, and approximates

$$\int f(v) \varphi(v) dv \approx \frac{1}{R} \sum_{r=1}^R f(v_r). \quad (141)$$

For normal random variables, Gauss-Hermite quadrature uses

$$\int_{-\infty}^{\infty} f(x) e^{-x^2} dx \approx \sum_{r=1}^R w_r f(x_r), \quad (142)$$

where x_r and w_r are nodes and weights. If $Y \sim N(\mu, \sigma^2)$, then with $Y = \mu + \sqrt{2}\sigma z$,

$$\mathbb{E}[f(Y)] \approx \pi^{-1/2} \sum_{r=1}^R w_r f(\mu + \sqrt{2}\sigma x_r). \quad (143)$$

3.15 Nested Logit Choice Probability and Decomposition

Suppose product j belongs to nest $g(j)$. Let

$$I_g = \log \sum_{k \in g} \exp(\delta_k / \lambda_g), \quad 0 < \lambda_g \leq 1. \quad (144)$$

The nested logit probability is

$$s_j = \underbrace{\frac{\exp(\delta_j/\lambda_g)}{\sum_{k \in g} \exp(\delta_k/\lambda_g)}}_{s_{j|g}} \underbrace{\frac{\exp(\lambda_g I_g)}{\sum_h \exp(\lambda_h I_h)}}_{s_g}. \quad (145)$$

The first term is the probability of choosing product j conditional on choosing nest g . The second term is the probability of choosing nest g . Products in the same nest share shocks and therefore substitute more closely with one another.

3.16 Nested Logit Linearization

Let the outside good be the only product in group 0, with $\delta_0 = 0$ and share s_0 . We have

$$\log s_j = \log s_{j|g} + \log s_g. \quad (146)$$

Using the formulas above,

$$\log s_{j|g} = \frac{\delta_j}{\lambda_g} - I_g, \quad (147)$$

$$\log s_g = \lambda_g I_g - \log \left(\sum_h \exp(\lambda_h I_h) \right), \quad (148)$$

$$\log s_0 = -\log \left(\sum_h \exp(\lambda_h I_h) \right). \quad (149)$$

Therefore

$$\log s_j - \log s_0 = \frac{\delta_j}{\lambda_g} - (1 - \lambda_g) I_g. \quad (150)$$

Since

$$\log s_{j|g} = \frac{\delta_j}{\lambda_g} - I_g, \quad (151)$$

rearrange to obtain the common linear equation

$$\log s_j - \log s_0 = \delta_j + (1 - \lambda_g) \log s_{j|g}. \quad (152)$$

Both price and the within-nest share are endogenous when unobserved product quality is present, so instruments are needed.

3.17 BLP Mean-Utility Inversion

In a random-coefficients demand model, write utility as

$$u_{ijt} = \delta_{jt} + \mu_{ijt}(\theta_2) + \epsilon_{ijt}, \quad (153)$$

where

$$\delta_{jt} = x'_{jt}\beta - \alpha p_{jt} + \zeta_{jt} \quad (154)$$

is the mean utility and μ_{ijt} is the random-coefficient deviation. Given nonlinear parameters θ_2 , the model share is

$$\sigma_{jt}(\delta_t, \theta_2) = \int \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{k \in \mathcal{J}_t} \exp(\delta_{kt} + \mu_{ikt})} dF_i. \quad (155)$$

Observed shares s_{jt} solve

$$s_{jt} = \sigma_{jt}(\delta_t, \theta_2), \quad j \in \mathcal{J}_t. \quad (156)$$

BLP computes $\delta_t(\theta_2)$ by the contraction mapping

$$\delta_t^{(r+1)} = \delta_t^{(r)} + \log s_t - \log \sigma_t(\delta_t^{(r)}, \theta_2). \quad (157)$$

At a fixed point,

$$\delta_t^* = \delta_t^* + \log s_t - \log \sigma_t(\delta_t^*, \theta_2), \quad (158)$$

so $s_t = \sigma_t(\delta_t^*, \theta_2)$. The contraction proves existence, uniqueness, and a practical algorithm.

3.18 BLP GMM and Concentrating Out Linear Parameters

Given $\delta(\theta_2)$, the unobserved product characteristic is

$$\zeta_{jt}(\theta) = \delta_{jt}(\theta_2) - x'_{jt}\beta + \alpha p_{jt}. \quad (159)$$

Let X_1 collect the linear regressors $(x'_{jt}, -p_{jt})$ and let $\theta_1 = (\beta', \alpha)'$ with the sign convention $\delta = X_1\theta_1 + \zeta$. Let Z be instruments. The GMM objective is

$$Q(\theta) = \zeta(\theta)'ZWZ'\zeta(\theta). \quad (160)$$

For fixed θ_2 , $\delta(\theta_2)$ is fixed, so the optimal linear parameters solve weighted IV:

$$\hat{\theta}_1(\theta_2) = (X_1'ZWZ'X_1)^{-1} X_1'ZWZ'\delta(\theta_2). \quad (161)$$

Thus the numerical search can be over nonlinear parameters only.

3.19 Control Function for Price Endogeneity

Suppose price is determined by observed variables and an anticipated demand/cost shock,

$$p_{jt} = \Pi(x_{jt}, w_{jt}) + v_{jt}. \quad (162)$$

If unobserved product quality satisfies

$$\zeta_{jt} = \rho v_{jt} + \eta_{jt}, \quad \mathbb{E}[\eta_{jt} \mid x_{jt}, w_{jt}, v_{jt}] = 0, \quad (163)$$

then the first-stage residual $\hat{v}_{jt} = p_{jt} - \hat{\Pi}(x_{jt}, w_{jt})$ is a control for price endogeneity. The demand utility includes $\rho\hat{v}_{jt}$, so the remaining unobservable is orthogonal to price conditional on the control.

4 Chapter 5: Merger Analysis

4.1 Conduct Parameter in a Homogeneous-Goods Market

Let $P(Q)$ be inverse demand and $Q = \sum_f q_f$. A conduct parameter representation of marginal revenue is

$$MR(Q) = P(Q) + \lambda QP'(Q). \quad (164)$$

The FOC is

$$P(Q) + \lambda QP'(Q) = MC(q_f). \quad (165)$$

This nests benchmark cases:

- Bertrand/perfect competition in a homogeneous-good price-taking approximation: $\lambda = 0$.
- Cournot: firm f has $MR_f = P(Q) + q_f P'(Q)$, so $\lambda = q_f/Q = s_f$.
- Joint-profit maximization: $MR = P(Q) + QP'(Q)$, so $\lambda = 1$.

4.2 Linear Demand, Linear Cost, and Nonidentification of Conduct

Let inverse demand be

$$P^D(Q) = \frac{\alpha_0}{\alpha_1} + \frac{1}{\alpha_1}Q + \frac{\alpha_2}{\alpha_1}X + \frac{1}{\alpha_1}u^D, \quad (166)$$

and marginal cost be

$$MC(q_f) = \beta_0 + \beta_1 q_f + \beta_2 W + u^S. \quad (167)$$

The conduct FOC is

$$P^S(Q) + \frac{\lambda}{\alpha_1}Q = \beta_0 + \beta_1 q_f + \beta_2 W + u^S. \quad (168)$$

If firms are symmetric, $q_f = Q/N$. Then

$$P^S(Q) = \beta_0 + \left(\frac{\beta_1}{N} - \frac{\lambda}{\alpha_1} \right) Q + \beta_2 W + u^S. \quad (169)$$

Let

$$\gamma = \frac{\beta_1}{N} - \frac{\lambda}{\alpha_1}. \quad (170)$$

Demand shifters and cost shifters can identify α_1 and γ , but not β_1 and λ separately. High price at a given quantity could be due to high marginal cost or high market power.

4.3 Identification with Cost Data

If reliable cost data identify $MC(q_f)$ directly, then β_1 is known. Since γ and α_1 are identified from the demand and pricing equations,

$$\lambda = \alpha_1 \left(\frac{\beta_1}{N} - \gamma \right). \quad (171)$$

The cost data break the markup-cost observational equivalence.

4.4 Identification with a Demand Rotator

A demand rotator changes the slope of inverse demand at a given quantity. Let

$$P^D(Q) = \frac{\alpha_0}{\alpha_1} + \frac{1}{\alpha_1}Q + \frac{\alpha_2}{\alpha_1}X + \frac{\alpha_3}{\alpha_1}QZ + \frac{1}{\alpha_1}u^D. \quad (172)$$

Then

$$\frac{\partial P^D(Q)}{\partial Q} = \frac{1 + \alpha_3 Z}{\alpha_1}. \quad (173)$$

The supply equation is

$$\begin{aligned} P^S(Q) &= \beta_0 + \frac{\beta_1}{N}Q + \beta_2 W - \lambda Q \frac{1 + \alpha_3 Z}{\alpha_1} + u^S \\ &= \beta_0 + \gamma_1 Q + \gamma_2 QZ + \beta_2 W + u^S, \end{aligned} \quad (174)$$

where

$$\gamma_1 = \frac{\beta_1}{N} - \frac{\lambda}{\alpha_1}, \quad \gamma_2 = -\frac{\lambda \alpha_3}{\alpha_1}. \quad (175)$$

If α_1 , α_3 , and γ_2 are identified, then

$$\lambda = -\frac{\gamma_2 \alpha_1}{\alpha_3}. \quad (176)$$

4.5 Differentiated Products: Conduct Matrix

For two products, the generalized FOCs are

$$Q_1(p) + (p_1 - c_1) \frac{\partial Q_1(p)}{\partial p_1} + \Delta_{12}(p_2 - c_2) \frac{\partial Q_2(p)}{\partial p_1} = 0, \quad (177)$$

$$Q_2(p) + (p_2 - c_2) \frac{\partial Q_2(p)}{\partial p_2} + \Delta_{21}(p_1 - c_1) \frac{\partial Q_1(p)}{\partial p_2} = 0. \quad (178)$$

If $\Delta_{12} = \Delta_{21} = 0$, firms price independently. If both equal one, the products' profits are internalized jointly. Identification requires variation that changes the relevant cross-derivatives without being absorbed by cost.

4.6 Pre-Merger Bertrand FOCs and Marginal Cost Recovery

There are J products. Let $\Delta_{jk}^{pre} = 1$ if products j and k are owned by the same pre-merger firm and zero otherwise. For product k ,

$$s_k(p) + \sum_{j=1}^J \Delta_{jk}^{pre} (p_j - mc_j) \frac{\partial s_j(p)}{\partial p_k} = 0. \quad (179)$$

Define the matrix $\Omega^{pre}(p)$ by

$$\Omega_{jk}^{pre}(p) = -\Delta_{jk}^{pre} \frac{\partial s_j(p)}{\partial p_k}. \quad (180)$$

Stacking FOCs gives

$$s(p) = \Omega^{pre}(p)(p - mc). \quad (181)$$

Thus

$$mc = p - [\Omega^{pre}(p)]^{-1} s(p). \quad (182)$$

4.7 Post-Merger Simulation

A merger changes ownership from Δ^{pre} to Δ^{post} , hence Ω^{pre} to Ω^{post} . Holding demand and marginal costs fixed, post-merger prices solve

$$p^{post} = mc + [\Omega^{post}(p^{post})]^{-1} s(p^{post}). \quad (183)$$

This is a nonlinear system. A fixed-point algorithm starts from $p^{(0)}$, computes

$$p^{(r+1)} = mc + [\Omega^{post}(p^{(r)})]^{-1} s(p^{(r)}), \quad (184)$$

and iterates until convergence, or one solves the FOCs directly with a nonlinear equation solver.

4.8 Consumer Surplus in Logit and Mixed Logit

If indirect utility is linear in price,

$$U_{ij} = V_{ij} + \epsilon_{ij}, \quad V_{ij} = x_j \beta_i + \alpha_i p_j + \xi_j, \quad (185)$$

with $\alpha_i < 0$ when price lowers utility, the expected maximum utility is

$$\log \sum_{j=0}^J \exp(V_{ij}) + \gamma. \quad (186)$$

The compensating variation from pre to post prices is

$$CV_i = -\frac{1}{\alpha_i} \left[\log \sum_{j=0}^J \exp(V_{ij}^{post}) - \log \sum_{j=0}^J \exp(V_{ij}^{pre}) \right]. \quad (187)$$

If the notes use α_i as a positive marginal disutility and utility is $x_j\beta_i - \alpha_i p_j$, the equivalent formula is

$$CV_i = \frac{\log \sum_j \exp(V_{ij}^{post}) - \log \sum_j \exp(V_{ij}^{pre})}{\alpha_i}. \quad (188)$$

For mixed logit, average consumer surplus integrates CV_i over simulated consumer types.

5 Chapter 6: Entry and Exit

5.1 Monopoly Entry and Nonidentification

A potential entrant in market i enters if

$$\pi(x_i, F_i) = v(x_i) - F_i \geq 0, \quad (189)$$

so

$$F_i \leq v(x_i). \quad (190)$$

If F is independent of x with CDF Ψ , the entry probability is

$$p(x) = \mathbb{P}(\text{entry} \mid x) = \Psi(v(x)). \quad (191)$$

Without restrictions, v and Ψ are not separately identified. For any strictly increasing transformation T ,

$$F \leq v(x) \iff T(F) \leq T(v(x)). \quad (192)$$

Thus the same entry probability can be generated by transformed fixed costs and transformed variable profits.

5.2 Identification with Known Distribution or Known Profit

If Ψ is known, then

$$v(x) = \Psi^{-1}(p(x)). \quad (193)$$

If v is known from demand and cost estimates, then for $z = v(x)$,

$$\Psi(z) = \mathbb{P}(F \leq z) = \mathbb{P}(F \leq v(x)) = p(x), \quad (194)$$

so Ψ is identified over the support of $v(x)$.

5.3 Identification under Homogeneity of Variable Profits

Suppose $x = (z, \tilde{x})$ and

$$v(z, \tilde{x}) = zv(1, \tilde{x}), \quad (195)$$

with normalization $v(1, \tilde{x}_0) = 1$. Then

$$\Psi(z) = \mathbb{P}(F \leq z) = \mathbb{P}(F \leq zv(1, \tilde{x}_0)) = p(z, \tilde{x}_0). \quad (196)$$

Once Ψ is identified, for any $(z, \tilde{x})$,

$$zv(1, \tilde{x}) = \Psi^{-1}(p(z, \tilde{x})), \quad (197)$$

so

$$v(1, \tilde{x}) = \frac{\Psi^{-1}(p(z, \tilde{x}))}{z}. \quad (198)$$

5.4 Complete-Information Homogeneous Oligopoly Entry

Let y_m be the equilibrium number of firms in market m . If there are y firms, per-firm profit is

$$\pi(y, x, F) = v_y(x) - F. \quad (199)$$

The equilibrium number y is characterized by

$$v_y(x) \geq F, \quad v_{y+1}(x) < F. \quad (200)$$

Hence

$$\mathbb{P}(Y = y \mid x) = \Psi(v_y(x)) - \Psi(v_{y+1}(x)). \quad (201)$$

Also,

$$\mathbb{P}(Y \geq y \mid x) = \Psi(v_y(x)). \quad (202)$$

The identification logic for each v_y is the same as in the monopoly-entry case.

5.5 Ordered-Response Likelihood

For observations (y_m, x_m) , the log likelihood is

$$\ell(v, \Psi) = \sum_{m=1}^M \log [\Psi(v_{y_m}(x_m)) - \Psi(v_{y_m+1}(x_m))]. \quad (203)$$

If Ψ is the normal CDF and $v_y(x)$ is parameterized by thresholds and covariates, this is an ordered probit-type likelihood.

5.6 Two-Firm Complete-Information Entry Game

There are two firms i and j . Firm i 's profit is

$$\pi_i(y_j, x_i, f_i) = v_{0i}(x_i) + y_j v_{1i}(x_i) - f_i, \quad (204)$$

where $v_{1i}(x_i) \leq 0$ is the competitive effect of firm j on firm i . Pure-strategy Nash equilibrium conditions are:

$$(0, 0) : \quad f_i \geq v_{0i}(x_i), \quad f_j \geq v_{0j}(x_j), \quad (205)$$

$$(1, 1) : f_i \leq v_{0i}(x_i) + v_{1i}(x_i), \quad f_j \leq v_{0j}(x_j) + v_{1j}(x_j), \quad (206)$$

$$(1, 0) : f_i \leq v_{0i}(x_i), \quad f_j \geq v_{0j}(x_j) + v_{1j}(x_j), \quad (207)$$

$$(0, 1) : f_i \geq v_{0i}(x_i) + v_{1i}(x_i), \quad f_j \leq v_{0j}(x_j). \quad (208)$$

Because $v_{1i}, v_{1j} \leq 0$, there is generally a rectangle in (f_i, f_j) where both $(1, 0)$ and $(0, 1)$ are equilibria. This is the multiple-equilibrium problem.

5.7 Moment Inequalities with Multiple Equilibria

When a unique equilibrium selection rule is not specified, some outcomes are only bounded. Let $H(1, 0 | x)$ be the probability of cost draws for which $(1, 0)$ is the unique equilibrium, and let $M(x)$ be the probability of the multiple-equilibrium region where either $(1, 0)$ or $(0, 1)$ may be selected. Then

$$H(1, 0 | x) \leq \mathbb{P}(1, 0 | x) \leq H(1, 0 | x) + M(x). \quad (209)$$

Similarly,

$$H(0, 1 | x) \leq \mathbb{P}(0, 1 | x) \leq H(0, 1 | x) + M(x). \quad (210)$$

The model imposes moment inequalities rather than equalities. A GMM criterion can penalize violations of lower and upper bounds.

5.8 Incomplete-Information Entry Thresholds

In incomplete information, firm i observes f_i but not f_j . A threshold strategy is

$$y_i = \mathbf{1}\{f_i \leq t_i\}, \quad y_j = \mathbf{1}\{f_j \leq t_j\}. \quad (211)$$

If firm i believes firm j enters with probability $G_j(t_j)$, the expected payoff from entering is

$$\begin{aligned} & [v_{0i}(x_i) - f_i][1 - G_j(t_j)] + [v_{0i}(x_i) + v_{1i}(x_i) - f_i]G_j(t_j) \\ & = v_{0i}(x_i) + v_{1i}(x_i)G_j(t_j) - f_i. \end{aligned} \quad (212)$$

The threshold makes the entrant indifferent:

$$t_i = v_{0i}(x_i) + v_{1i}(x_i)G_j(t_j). \quad (213)$$

Similarly,

$$t_j = v_{0j}(x_j) + v_{1j}(x_j)G_i(t_i). \quad (214)$$

This system can have multiple solutions, so equilibrium selection can again matter.

6 Chapter 7: Dynamic Decision Model

6.1 Bellman Equation for a Single Agent

There are states $s \in S$, actions $a \in A = \{0, 1, \dots, K\}$, transition probabilities $g(a, s, s')$, discount factor $\beta < 1$, and per-period payoff

$$\pi(a, s) + \epsilon_a. \quad (215)$$

The choice-specific value is

$$V(a, s, \epsilon) = \pi(a, s) + \epsilon_a + \beta \sum_{s'} g(a, s, s') V(s'), \quad (216)$$

where $V(s')$ is the ex-ante value before next period's shocks are realized. Define the choice-specific mean value

$$v(a, s) = \pi(a, s) + \beta \sum_{s'} g(a, s, s') V(s'). \quad (217)$$

The optimal action satisfies

$$a \in \arg \max_{k \in A} \{v(k, s) + \epsilon_k\}. \quad (218)$$

6.2 Logit CCPs and Inclusive-Value Bellman Operator

If ϵ_k are i.i.d. type-I extreme value, then

$$p(a | s) = \frac{\exp(v(a, s))}{\sum_{k \in A} \exp(v(k, s))}. \quad (219)$$

The ex-ante value is

$$\begin{aligned} V(s) &= \mathbb{E} \left[\max_{a \in A} \{v(a, s) + \epsilon_a\} \right] \\ &= \log \left(\sum_{a \in A} \exp \left[\pi(a, s) + \beta \sum_{s'} g(a, s, s') V(s') \right] \right) + \gamma. \end{aligned} \quad (220)$$

This defines the Bellman operator Φ :

$$(\Phi V)(s) = \log \left(\sum_{a \in A} \exp \left[\pi(a, s) + \beta \sum_{s'} g(a, s, s') V(s') \right] \right) + \gamma. \quad (221)$$

Because $\beta < 1$, the operator is a contraction under the sup norm, so value iteration converges to the unique fixed point.

6.3 Identification Logic

The primitives are (Π, F, β, G) . The transition G is identified from observed (s_t, a_t, s_{t+1}) . The CCPs $p(a | s)$ are identified from observed choices conditional on state. However, β and Π are not jointly identified without restrictions: a high probability of investment can be explained by either high future value (large β) or low current investment cost. Standard practice fixes β and F , normalizes one action payoff such as $\pi(0, s) = 0$, and identifies payoff differences.

6.4 Nested Fixed-Point Estimation

For each parameter vector θ , solve the dynamic program:

$$V_\theta^{(r+1)} = \Phi_\theta(V_\theta^{(r)}). \quad (222)$$

Let V_θ^* be the fixed point. Compute implied CCPs

$$p_\theta(a | s) = \frac{\exp [\pi_\theta(a, s) + \beta \sum_{s'} g_\theta(a, s, s') V_\theta^*(s')]}{\sum_k \exp [\pi_\theta(k, s) + \beta \sum_{s'} g_\theta(k, s, s') V_\theta^*(s')]} \quad (223)$$

The likelihood for data (a_t, s_t, s_{t+1}) is

$$L(\theta) = \prod_t p_\theta(a_t | s_t) g_\theta(s_{t+1} | s_t, a_t), \quad (224)$$

with log likelihood

$$\ell(\theta) = \sum_t \log p_\theta(a_t | s_t) + \sum_t \log g_\theta(s_{t+1} | s_t, a_t). \quad (225)$$

The fixed point is nested inside the likelihood evaluation.

6.5 CCP Approach: Main Idea

The CCP approach estimates $\hat{p}(a | s)$ and $\hat{g}(s' | s, a)$ first. Then it uses the equilibrium condition

$$p = \Psi_\theta(p) \quad (226)$$

as a moment condition. Instead of solving a fixed point for every θ , evaluate the model's best response to the observed CCPs and choose θ to make it close to the observed CCPs:

$$\hat{\theta} = \arg \min_{\theta} [\hat{p} - \Psi_\theta(\hat{p})]' W [\hat{p} - \Psi_\theta(\hat{p})]. \quad (227)$$

6.6 Expected Shock Conditional on Choice under Logit

The CCP value formula requires $\mathbb{E}[\epsilon_a | a, s]$. Under logit,

$$\mathbb{E}[\epsilon_a | a, s] = \gamma - \log p(a | s). \quad (228)$$

Intuition: the realized shock conditional on choosing an unlikely action must be high, so $-\log p(a | s)$ increases as $p(a | s)$ falls.

A useful derivation uses the identity

$$\mathbb{E} \left[\max_k \{v_k + \epsilon_k\} \right] = \sum_a p_a (v_a + \mathbb{E}[\epsilon_a | a]) \quad (229)$$

and the inclusive value formula

$$\mathbb{E} \left[\max_k \{v_k + \epsilon_k\} \right] = \log \sum_k \exp(v_k) + \gamma. \quad (230)$$

Since $p_a = \exp(v_a) / \sum_k \exp(v_k)$,

$$v_a = \log p_a + \log \sum_k \exp(v_k). \quad (231)$$

Substituting into the first identity and canceling the common log-sum term yields

$$\sum_a p_a \mathbb{E}[\epsilon_a | a] = \gamma - \sum_a p_a \log p_a. \quad (232)$$

The logit distribution also implies the action-specific conditional mean $\gamma - \log p_a$, which satisfies the weighted identity above.

6.7 CCP Value Function in Matrix Form

Given CCPs $p(a | s)$ and transition g , the ex-ante value under policy p satisfies

$$\begin{aligned} V(s) &= \sum_a p(a | s) \left[\pi(a, s) + \mathbb{E}(\epsilon_a | a, s) + \beta \sum_{s'} g(a, s, s') V(s') \right] \\ &= \sum_a p(a | s) \left[\pi(a, s) + \gamma - \log p(a | s) + \beta \sum_{s'} g(a, s, s') V(s') \right]. \end{aligned} \quad (233)$$

Define $\Sigma(p)$ as the matrix that averages action-specific objects using $p(a | s)$ at each state, and G as the stacked transition matrix by action and state. Let Π be the stacked payoff vector and $E(p)$ the stacked vector $\gamma - \log p(a | s)$. Stacking (233) across states yields

$$V = \Sigma(p)[\Pi + E(p)] + \beta \Sigma(p)GV. \quad (234)$$

Therefore

$$V = [I - \beta \Sigma(p)G]^{-1} \Sigma(p)[\Pi + E(p)]. \quad (235)$$

This is the key Hotz-Miller computational simplification.

6.8 CCP Optimality Equation

Given V from (235), the model-implied best-response CCP is

$$\Psi_{\theta}(p)(a | s) = \frac{\exp [\pi_{\theta}(a, s) + \beta \sum_{s'} g(a, s, s') V(s')]}{\sum_k \exp [\pi_{\theta}(k, s) + \beta \sum_{s'} g(k, s, s') V(s')]} \quad (236)$$

At the true parameter and equilibrium CCP, $p = \Psi_{\theta}(p)$.

6.9 Finite-Mixture Identification: Factorization

Suppose agents have unobserved type $m = 1, \dots, M$ with probabilities π^m . In the simplified binary-action setup, define

$$\lambda_{\xi}^m = P^m(a = 1 | x = \xi), \quad \lambda_{\xi}^{*m} = p^{*m}(a_1 = 1, x_1 = \xi). \quad (237)$$

For selected covariate values $\xi_1, \dots, \xi_{M-1}$ define

$$L = \begin{pmatrix} 1 & \lambda_{\xi_1}^1 & \dots & \lambda_{\xi_{M-1}}^1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \lambda_{\xi_1}^M & \dots & \lambda_{\xi_{M-1}}^M \end{pmatrix}, \quad V = \text{diag}(\pi^1, \dots, \pi^M), \quad (238)$$

and

$$D_k^* = \text{diag}(\lambda_k^{*1}, \dots, \lambda_k^{*M}). \quad (239)$$

Observable probability matrices satisfy

$$P = L'VL, \quad P_k^* = L'D_k^*VL. \quad (240)$$

If P is nonsingular,

$$\begin{aligned} P^{-1}P_k^* &= (L'VL)^{-1}L'D_k^*VL \\ &= L^{-1}V^{-1}(L')^{-1}L'D_k^*VL \\ &= L^{-1}V^{-1}D_k^*VL \\ &= L^{-1}D_k^*L. \end{aligned} \quad (241)$$

Thus the eigenvalues of $P^{-1}P_k^*$ identify the diagonal elements of D_k^* , and the eigenvectors identify L up to labeling. Then

$$V = (L')^{-1}PL^{-1} \quad (242)$$

identifies type probabilities.

6.10 EM Algorithm for Unobserved Dynamic States

Let the true state be (x_t, w_t) , where x_t is observed and w_t is hidden. The EM algorithm alternates:

1. **Expectation step.** Given current parameters, compute posterior probabilities

$$q_{it}^{(r+1)}(w) = \mathbb{P}(w_{it} = w \mid \{a_{it}, x_{it}\}_{t=1}^T; \theta^{(r)}). \quad (243)$$

These can be computed by forward-backward recursions in hidden Markov models.

2. **Maximization step.** Treat $q_{it}^{(r+1)}(w)$ as fractional type/state membership weights and update payoff, transition, initial-distribution, and CCP parameters.

The algorithm avoids directly summing over all hidden-state sequences in every maximization step.

7 Chapter 8: Dynamic Game

7.1 Multi-Agent Value Function

There are players $i = 1, \dots, N$, action profile $a = (a_i, a_{-i})$, state s , and payoff

$$\pi_i(a, s) + \epsilon_{ia_i}. \quad (244)$$

Let player i believe that action profiles are drawn according to $\sigma_i(a \mid s)$. The ex-ante value under this belief satisfies

$$\begin{aligned} V_i(\sigma_i, s) &= \sum_a \sigma_i(a \mid s) \left[\pi_i(a, s) + \beta \sum_{s'} g(a, s, s') V_i(\sigma_i, s') \right] \\ &\quad + \sum_{k=0}^K \mathbb{E}[\epsilon_{ik} \mid a_i = k, s] \sigma_i(a_i = k \mid s). \end{aligned} \quad (245)$$

Stacking across states,

$$V_i(\sigma_i) = [I - \beta \Sigma_i(\sigma_i) G]^{-1} [\Sigma_i(\sigma_i) \Pi_i + D_i(\sigma_i)]. \quad (246)$$

7.2 Choice-Specific Value and Equilibrium CCP

Given beliefs about rivals, player i 's choice-specific mean value is

$$v_i(\sigma_i, a_i, s) = \sum_{a_{-i}} \sigma_i(a_{-i} \mid s) \left[\pi_i(a_i, a_{-i}, s) + \beta \sum_{s'} g(a_i, a_{-i}, s, s') V_i(\sigma_i, s') \right]. \quad (247)$$

The optimal CCP is

$$p_i(a_i \mid s) = \mathbb{P}\{v_i(\sigma_i, a_i, s) + \epsilon_{ia_i} \geq v_i(\sigma_i, a'_i, s) + \epsilon_{ia'_i} \forall a'_i\}. \quad (248)$$

If shocks are logit,

$$p_i(a_i \mid s) = \frac{\exp(v_i(\sigma_i, a_i, s))}{\sum_{k \in A_i} \exp(v_i(\sigma_i, k, s))}. \quad (249)$$

In a Markov perfect equilibrium, beliefs are consistent with strategies. Therefore the joint CCP profile satisfies

$$p = \Psi(p). \quad (250)$$

7.3 Identification Order Condition

The payoff vector Π_i generally has one payoff for each action profile and state, so its dimension is large. The CCP equilibrium restrictions provide roughly one restriction for each non-normalized own action in each state. Normalizing $\pi_i(0, a_{-i}, s) = 0$ is often not enough. A common restriction is that player i 's payoff does not directly depend on rivals' states:

$$\pi_i(a, s_i, s_{-i}) = \pi_i(a, s_i, s'_{-i}) \quad \forall s_{-i}, s'_{-i}. \quad (251)$$

This reduces dimensionality and can make the order condition plausible.

7.4 CCP Estimation for Dynamic Games

Estimate $\hat{p}$ and $\hat{G}$ directly from observed states and actions. For each candidate payoff parameter θ_1 , compute

$$V_i(\hat{p}; \theta_1) = [I - \beta \Sigma_i(\hat{p}) \hat{G}]^{-1} [\Sigma_i(\hat{p}) \Pi_i(\theta_1) + D_i(\hat{p})]. \quad (252)$$

Then compute $\Psi_{\theta_1, \hat{G}}(\hat{p})$ using the choice-specific value functions. Estimate

$$\hat{\theta}_1 = \arg \min_{\theta_1} \left[\hat{p} - \Psi_{\theta_1, \hat{G}}(\hat{p}) \right]' W \left[\hat{p} - \Psi_{\theta_1, \hat{G}}(\hat{p}) \right]. \quad (253)$$

This avoids solving for all equilibria, which is crucial because dynamic games may have multiple Markov perfect equilibria.

7.5 Continuous Controls: Inverting the Policy Function

Let a player's action be (a_{id}, a_{ic}) , where a_{id} is discrete and a_{ic} is continuous. Suppose the payoff includes

$$\epsilon_{ic} a_{ic} \quad (254)$$

and the policy $a_{ic}(s, \epsilon_{ic})$ is strictly increasing in ϵ_{ic} . If $F_{a_c}(a | s)$ is the observed conditional distribution of continuous actions and $F_{\epsilon_c}(\epsilon | s)$ is known, then

$$\begin{aligned} F_{a_c}(a | s) &= \mathbb{P}(a_c(s, \epsilon_c) \leq a | s) \\ &= \mathbb{P}(\epsilon_c \leq a_c^{-1}(s, a) | s) \\ &= F_{\epsilon_c}(a_c^{-1}(s, a) | s). \end{aligned} \quad (255)$$

Hence

$$a_c^{-1}(s, a) = F_{\epsilon_c}^{-1}(F_{a_c}(a | s) | s), \quad (256)$$

and the policy can be represented by quantile matching:

$$a_c(s, \epsilon_c) = F_{a_c}^{-1}(F_{\epsilon_c}(\epsilon_c | s) | s). \quad (257)$$

7.6 BBL Inequality Estimator

Let a_i^* be the observed equilibrium strategy and a_i' an alternative strategy. In equilibrium,

$$V_i(a_i^*, a_{-i}^*; s) \geq V_i(a_i', a_{-i}^*; s) \quad (258)$$

for all i and s . This leads to an objective function that penalizes violations:

$$Q(\theta) = \sum_s \sum_i \sum_{a_i'} [\min\{V_i(a_i^*, a_{-i}^*; s; \theta) - V_i(a_i', a_{-i}^*; s; \theta), 0\}]^2. \quad (259)$$

The value functions are often approximated by forward simulation under the estimated policy and perturbed policies.

7.7 Dynamic Oligopoly with Entry, Exit, and Investment

In a dynamic oligopoly, state s_{it} may be firm capacity. Incumbents decide whether to exit and how much to invest; potential entrants decide whether to enter. A generic period profit is

$$\pi_i(s, a; \theta) = \tilde{\pi}_i(s; \alpha, \delta) - \Gamma(x_i; \gamma_i) + \Phi_i(a_i; \kappa_i, \phi_i), \quad (260)$$

where $\tilde{\pi}_i$ is product-market profit, Γ is investment/disinvestment cost, and Φ captures entry costs or exit scrap values. The state transition maps current capacity and investment into future capacity. Counterfactual regulation changes parameters such as entry or investment costs and then recomputes equilibrium behavior.

7.8 Oblivious Equilibrium

In industries with many firms, a full Markov strategy depending on every rival's state is infeasible. Oblivious equilibrium assumes each firm conditions only on its own state and the long-run average industry state $\tilde{s}$, not the exact current vector of rivals' states.

If firms use an oblivious strategy μ and the entry rate is λ , a firm's state follows a Markov chain with transition probabilities $P_\mu^k(x^e, x)$. The long-run expected number of firms in state x is

$$\tilde{s}_{\mu, \lambda}(x) = \lambda \sum_{k=0}^{\infty} P_\mu^k(x^e, x). \quad (261)$$

A firm evaluates profits using $\pi(x, \tilde{s}_{\mu, \lambda})$. An oblivious equilibrium is a strategy and entry rate such that firms optimize against this long-run average state and the free-entry condition holds:

$$\lambda (\beta \mathbb{E}[V(x^e | \mu, \lambda)] - \kappa) = 0, \quad \beta \mathbb{E}[V(x^e | \mu, \lambda)] - \kappa \leq 0, \quad \lambda \geq 0. \quad (262)$$

8 Chapter 9: Auctions

8.1 Second-Price/Button Auction under Symmetric Information

Let bidder i observe signal $X_i = x$. Let Y_1 be the highest rival signal. Define

$$w(x, y) = \mathbb{E}[u(V, x) | X_i = x, Y_1 = y]. \quad (263)$$

Under affiliation, $w(x, y)$ is increasing in x .

In a button auction where dropout prices are not observed, guess a symmetric dropout rule

$$b(x) = w(x, x). \quad (264)$$

If the current price is b and a bidder wins at that price, the strongest rival's inferred signal is $y = b^{-1}(b)$. The expected payoff from staying until price b is

$$w(x, y) - b = w(x, y) - w(y, y). \quad (265)$$

This is positive iff $x > y$. Therefore the optimal dropout price is the point where $y = x$, namely $b(x) = w(x, x)$.

In the private-values case, $w(x, y) = x$, so the equilibrium bid/dropout is simply

$$b(x) = x. \quad (266)$$

8.2 Second-Price Auction with Reserve Price

Let the reserve price be r . In private values, a bidder participates iff $x \geq r$ and bids x . In common values, the participation threshold solves

$$x^*(r) = \inf \{x : \mathbb{E}[w(x, Y_1) \mid X = x, Y_1 < x] \geq r\}. \quad (267)$$

The conditioning $Y_1 < x$ reflects the event that a bidder with signal x wins against rivals who remain below x .

8.3 IPV Button Auction with Winning Bid Only

In an IPV second-price auction with n_t potential bidders and reserve r_t , the winning price is

$$w_t = \max\{X_{2:n_t}, r_t\}, \quad (268)$$

where $X_{2:n}$ is the second-highest valuation. If no bidder has $X_i \geq r_t$, then no sale occurs. The probability of no actual bidder is

$$\mathbb{P}(m_t = 0) = F_X(r_t)^{n_t}. \quad (269)$$

The probability exactly one bidder exceeds the reserve is

$$\mathbb{P}(m_t = 1) = n_t[1 - F_X(r_t)]F_X(r_t)^{n_t-1}. \quad (270)$$

If at least two bidders participate and the observed winning bid is w , then w is the second-highest order statistic. Its density is

$$h_t(w) = n_t(n_t - 1)F_X(w)^{n_t-2}[1 - F_X(w)]f_X(w). \quad (271)$$

Conditional on observing auctions with at least one participant, the likelihood contribution divides by $1 - F_X(r_t)^{n_t}$.

8.4 Optimal Reserve Price in IPV Second-Price Auction

Let the seller value the item at x_0 . Expected revenue with reserve r is

$$R(r) = x_0 F(r)^n + r n F(r)^{n-1} [1 - F(r)] + \int_r^{\bar{x}} w n(n-1) F(w)^{n-2} [1 - F(w)] f(w) dw. \quad (272)$$

Differentiating and simplifying gives the classic first-order condition

$$r = x_0 + \frac{1 - F(r)}{f(r)}. \quad (273)$$

The reserve exceeds the seller's own value by the inverse hazard rate.

8.5 First-Price IPV Auction: Equilibrium Differential Equation

In a first-price sealed-bid auction, the highest bidder wins and pays her own bid. Assume IPV, symmetric increasing strategy $b = \beta(x)$, inverse $x = \eta(b)$, and reserve r with $\beta(r) = r$. If a bidder with value x bids b , expected profit is

$$\pi(b, x) = (x - b) F_X(\eta(b))^{n-1}. \quad (274)$$

The FOC with respect to b is

$$-F_X(\eta(b))^{n-1} + (x - b)(n-1)F_X(\eta(b))^{n-2} f_X(\eta(b)) \eta'(b) = 0. \quad (275)$$

At a symmetric equilibrium, $b = \beta(x)$ and $\eta(b) = x$. Since $\eta'(\beta(x)) = 1/\beta'(x)$,

$$-F_X(x)^{n-1} + (x - \beta(x))(n-1)F_X(x)^{n-2} f_X(x) \frac{1}{\beta'(x)} = 0. \quad (276)$$

Rearrange:

$$\beta'(x) = [x - \beta(x)] \frac{(n-1)f_X(x)}{F_X(x)}. \quad (277)$$

Let $G(x) = F_X(x)^{n-1}$ and $g(x) = G'(x) = (n-1)F_X(x)^{n-2} f_X(x)$. Then

$$[x - \beta(x)]g(x) - \beta'(x)G(x) = 0. \quad (278)$$

Equivalently,

$$\beta'(x) + \frac{g(x)}{G(x)}\beta(x) = x \frac{g(x)}{G(x)}. \quad (279)$$

8.6 Solving the First-Price Differential Equation

The differential equation is linear:

$$\beta'(x) + p(x)\beta(x) = q(x), \quad p(x) = \frac{g(x)}{G(x)}, \quad q(x) = x \frac{g(x)}{G(x)}. \quad (280)$$

An integrating factor $\mu(x)$ satisfies

$$\mu'(x) = \mu(x)p(x). \quad (281)$$

Thus

$$\mu(x) = \exp \left(\int_r^x \frac{g(t)}{G(t)} dt \right) = \exp (\log G(x) - \log G(r)). \quad (282)$$

Set the arbitrary constant so that $\mu(x) = G(x)$. Multiplying the differential equation by $G(x)$ gives

$$[G(x)\beta(x)]' = xg(x). \quad (283)$$

Integrate from r to x :

$$G(x)\beta(x) - G(r)\beta(r) = \int_r^x tg(t)dt. \quad (284)$$

Using $\beta(r) = r$,

$$\beta(x) = \frac{rG(r) + \int_r^x tg(t)dt}{G(x)}. \quad (285)$$

Integrate by parts:

$$\int_r^x tg(t)dt = xG(x) - rG(r) - \int_r^x G(t)dt. \quad (286)$$

Therefore

$$\beta(x) = x - \frac{\int_r^x G(t)dt}{G(x)} = x - \frac{\int_r^x F_X(t)^{n-1}dt}{F_X(x)^{n-1}}. \quad (287)$$

The second term is the bid markdown below value. It shrinks as competition increases.

8.7 First-Price MLE with Winning Bids

If only winning bids w_t are observed, the density of the highest bid is obtained by change of variables from the highest valuation:

$$h_t(w) = n_t F_X(\eta_t(w))^{n_t-1} f_X(\eta_t(w)) \eta_t'(w). \quad (288)$$

Using the first-order condition,

$$\eta_t'(w) = \frac{F_X(\eta_t(w))}{(n_t - 1)f_X(\eta_t(w))[\eta_t(w) - w]}. \quad (289)$$

Thus

$$h_t(w) = \frac{n_t F_X(\eta_t(w))^{n_t}}{(n_t - 1)[\eta_t(w) - w]}. \quad (290)$$

Conditional on sale, the likelihood contribution is

$$\frac{h_t(w_t)}{1 - F_X(r_t)^{n_t}}. \quad (291)$$

8.8 GPV Nonparametric Inversion

With all bids observed and fixed number of bidders n , let F_B and f_B be the bid distribution and density. The distribution of the highest rival bid is

$$H(b) = F_B(b)^{n-1}, \quad h(b) = (n-1)F_B(b)^{n-2}f_B(b). \quad (292)$$

A bidder with value x bidding b earns

$$\pi(b, x) = (x - b)H(b). \quad (293)$$

The FOC is

$$-H(b) + (x - b)h(b) = 0. \quad (294)$$

Thus the valuation corresponding to bid b is

$$x = b + \frac{H(b)}{h(b)} = b + \frac{F_B(b)}{(n-1)f_B(b)}. \quad (295)$$

The GPV estimator estimates F_B and f_B nonparametrically, forms pseudo-values

$$\hat{x}_{it} = b_{it} + \frac{\hat{F}_B(b_{it})}{(n-1)\hat{f}_B(b_{it})}, \quad (296)$$

and then estimates the valuation distribution from the pseudo-values.

8.9 Set Identification in English Auctions with Bid Increments

Actual English auctions may have jump bids, missed dropout observations, and minimum bid increments Δ . Haile-Tamer style inequalities use two weak assumptions:

1. A bidder never bids above value: $b_i \leq x_i$.
2. The winning bid plus the bid increment exceeds the second-highest value: $x_{2:n} \leq w + \Delta$.

Let $G_{i:n}$ be the empirical distribution of the i th highest bid and $F_{i:n}$ the distribution of the i th highest value. The first assumption implies

$$F_{i:n}(x) \leq G_{i:n}(x), \quad (297)$$

while the second implies

$$F_{2:n}(x) \geq G_{1:n}(x + \Delta). \quad (298)$$

These inequalities bound the valuation distribution without fully specifying the auction's exact equilibrium.

A Supplementary Worked Derivations and Algebraic Details

This section adds slower algebraic steps and additional identification logic for the formulas above. It is meant to be read as a companion to the chapter-by-chapter derivations, especially when moving from an economic model to an estimating equation.

A.1 Chapter 3 Addendum: OLS Bias Written as an Omitted-Variable Bias

Return to

$$y = \beta_0 + \beta_l l + \beta_k k + \epsilon. \quad (299)$$

For clarity, partial out the constant and capital from labor and output. Let $\tilde{l}$ be the residual from projecting l on $(1, k)$, and let $\tilde{y}$ be the residual from projecting y on $(1, k)$. By the Frisch-Waugh-Lovell theorem, the OLS labor coefficient is

$$\hat{\beta}_l^{OLS} = \frac{\sum_i \tilde{l}_i \tilde{y}_i}{\sum_i \tilde{l}_i^2}. \quad (300)$$

The residualized outcome satisfies

$$\tilde{y}_i = \beta_l \tilde{l}_i + \tilde{\epsilon}_i, \quad (301)$$

where $\tilde{\epsilon}_i$ is the residual from projecting ϵ_i on $(1, k_i)$. Hence

$$\text{plim } \hat{\beta}_l^{OLS} = \beta_l + \frac{\mathbb{E}[\tilde{l}_i \tilde{\epsilon}_i]}{\mathbb{E}[\tilde{l}_i^2]}. \quad (302)$$

The static labor policy in (8) implies that, after controlling for k , labor still moves with productivity. Therefore $\mathbb{E}[\tilde{l}_i \tilde{\epsilon}_i] > 0$ under the usual monotone policy, giving upward labor bias. This makes precise the phrase “OLS attributes productivity to labor.”

A.2 Chapter 3 Addendum: Why OP Does Not Identify the Capital Coefficient in the First Stage

The OP first stage is

$$y_{jt} = \beta_l l_{jt} + \phi(k_{jt}, i_{jt}) + \eta_{jt}, \quad (303)$$

where

$$\phi(k, i) = \beta_0 + \beta_k k + h(k, i). \quad (304)$$

Even if ϕ is perfectly identified, one cannot separate $\beta_k k$ from $h(k, i)$ in the first stage. To see this, fix any candidate b_k . Define

$$h_{b_k}(k, i) = \phi(k, i) - \beta_0 - b_k k. \quad (305)$$

For every b_k , the same first-stage fit can be rationalized by a different h_{b_k} . The first stage therefore identifies only β_l and the composite ϕ , not β_k . The second stage uses the Markov restriction on productivity to restrict the possible decompositions of ϕ into $\beta_k k$ and productivity.

A.3 Chapter 3 Addendum: OP Second Stage as a Nonlinear Least-Squares Problem

After replacing β_l and ϕ by first-stage estimates, the second stage can be written as

$$\tilde{y}_{jt} = \beta_0 + \beta_k k_{jt} + g(\hat{\phi}_{j,t-1} - \beta_0 - \beta_k k_{j,t-1}) + e_{jt}, \quad (306)$$

where $\tilde{y}_{jt} = y_{jt} - \hat{\beta}_l l_{jt}$ and $e_{jt} = v_{jt} + \eta_{jt}$ plus first-stage approximation error. Approximate g by a polynomial,

$$g(\omega) = \sum_{r=0}^R \rho_r \omega^r. \quad (307)$$

Then the sample objective is

$$\min_{\beta_0, \beta_k, \rho} \sum_{j,t} \left[\tilde{y}_{jt} - \beta_0 - \beta_k k_{jt} - \sum_{r=0}^R \rho_r (\hat{\phi}_{j,t-1} - \beta_0 - \beta_k k_{j,t-1})^r \right]^2. \quad (308)$$

The problem is nonlinear in (β_0, β_k) because these parameters enter inside the polynomial argument. If instead one uses GMM, instruments can be functions of predetermined variables such as k_{jt} , $k_{j,t-1}$, and $i_{j,t-1}$.

A.4 Chapter 3 Addendum: Cost Heterogeneity Regression

Suppose a marginal cost equation has an observed shifter z and unobserved cost shock ϵ^c :

$$MC(q, z, \epsilon^c) = \tilde{C}'(q) + z'\gamma + \epsilon^c. \quad (309)$$

For a single-product monopolist, revealed marginal cost is

$$MC(D(p, x), z, \epsilon^c) = p + \frac{D(p, x)}{D_p(p, x)}. \quad (310)$$

Substitute the structural marginal cost:

$$p + \frac{D(p, x)}{D_p(p, x)} = \tilde{C}'(D(p, x)) + z'\gamma + \epsilon^c. \quad (311)$$

Move the known or parameterized non-shifter part to the left:

$$\underbrace{p + \frac{D(p, x)}{D_p(p, x)} - \tilde{C}'(D(p, x))}_{\text{revealed residual marginal cost}} = z'\gamma + \epsilon^c. \quad (312)$$

If $\mathbb{E}[\epsilon^c | z] = 0$, estimate by least squares. If z is endogenous, use instruments w satisfying

$$\mathbb{E} \left[w \left(p + \frac{D(p, x)}{D_p(p, x)} - \tilde{C}'(D(p, x)) - z'\gamma \right) \right] = 0. \quad (313)$$

A.5 Chapter 4 Addendum: Logit IIA and the Red-Bus/Blue-Bus Algebra

In multinomial logit,

$$\frac{s_j}{s_k} = \frac{\exp(\delta_j)}{\exp(\delta_k)} = \exp(\delta_j - \delta_k). \quad (314)$$

This odds ratio is independent of every other product. Suppose initially there are bus and train with equal mean utilities, so shares are

$$s_B = s_T = \frac{1}{2}. \quad (315)$$

Now split the bus alternative into two identical alternatives, blue bus and red bus, each with the same mean utility as the original bus and train. The logit shares become

$$s_{blue} = s_{red} = s_T = \frac{\exp(\delta)}{3 \exp(\delta)} = \frac{1}{3}. \quad (316)$$

The total bus share becomes 2/3, even though no real new transportation option was created. The problem is that the idiosyncratic shocks of blue and red bus are independent, so adding an identical bus creates an additional independent draw.

A.6 Chapter 4 Addendum: Logit Markups under Single-Product Bertrand

For single-product Bertrand,

$$mc_j = p_j + \frac{s_j}{\partial s_j / \partial p_j}. \quad (317)$$

With logit,

$$\frac{\partial s_j}{\partial p_j} = -\alpha s_j (1 - s_j). \quad (318)$$

Therefore

$$mc_j = p_j - \frac{1}{\alpha(1 - s_j)}, \quad p_j - mc_j = \frac{1}{\alpha(1 - s_j)}. \quad (319)$$

This formula illustrates why demand curvature and substitution patterns matter. If s_j is small, the markup is roughly $1/\alpha$. If s_j is large, the markup grows because a price increase loses fewer consumers to the outside option and rivals in the simple logit structure.

A.7 Chapter 4 Addendum: BLP Contraction Mapping Intuition

For a fixed θ_2 , define

$$T(\delta) = \delta + \log s - \log \sigma(\delta, \theta_2). \quad (320)$$

At the solution, $s = \sigma(\delta, \theta_2)$ and $T(\delta) = \delta$. If model shares are below data shares for product j , then $\log s_j - \log \sigma_j > 0$, so the update raises δ_j . If model shares are above data shares, the update lowers δ_j . The proof that T is a contraction uses two facts: model shares are increasing in own mean utility and satisfy adding-up with the outside good, so a uniform increase in all inside mean utilities cannot change all inside shares one-for-one. Iterating the map therefore moves model shares toward observed shares.

A.8 Chapter 4 Addendum: Instruments in Differentiated-Products Demand

In the equation

$$\delta_{jt} = x'_{jt}\beta - \alpha p_{jt} + \xi_{jt}, \quad (321)$$

price is correlated with ξ_{jt} because products with high unobserved quality optimally set high prices. Instruments must shift price while being uncorrelated with ξ_{jt} . Common categories include:

- cost shifters, such as input prices or transportation costs;
- characteristics of rival products, which affect competitive pricing incentives;
- functions of the number and location of products in characteristic space, such as BLP-style sums of rival characteristics.

The moment condition is

$$\mathbb{E}[Z_{jt}\xi_{jt}(\theta)] = 0. \quad (322)$$

Identification of random coefficients requires instruments that generate variation in substitution patterns, not only the average price level.

A.9 Chapter 5 Addendum: Multiproduct Bertrand in Matrix Form

Let $\mathcal{F}(j)$ denote the firm that owns product j . Firm f maximizes

$$\Pi_f(p) = \sum_{j:\mathcal{F}(j)=f} (p_j - mc_j)s_j(p)M, \quad (323)$$

where M is market size. The FOC for product k owned by firm f is

$$0 = s_k(p)M + \sum_{j:\mathcal{F}(j)=f} (p_j - mc_j)M \frac{\partial s_j(p)}{\partial p_k}. \quad (324)$$

Divide by M and use the ownership matrix $\Delta_{jk} = \mathbf{1}\{\mathcal{F}(j) = \mathcal{F}(k)\}$:

$$s_k(p) + \sum_j \Delta_{jk}(p_j - mc_j) \frac{\partial s_j(p)}{\partial p_k} = 0. \quad (325)$$

Define $\Omega_{jk} = -\Delta_{jk}\partial s_j/\partial p_k$. Then

$$s = \Omega(p - mc), \quad p - mc = \Omega^{-1}s. \quad (326)$$

This is the exact algebra behind the marginal-cost recovery formula. The merger changes only Δ if marginal costs and demand parameters are held fixed.

A.10 Chapter 5 Addendum: Logit Merger Matrix Elements

For simple logit,

$$\frac{\partial s_j}{\partial p_k} = \begin{cases} -\alpha s_j(1 - s_j), & j = k, \\ \alpha s_j s_k, & j \neq k. \end{cases} \quad (327)$$

Thus, for a pre-merger ownership matrix Δ^{pre} ,

$$\Omega_{jk}^{pre} = \begin{cases} \alpha s_j(1 - s_j), & j = k, \\ -\Delta_{jk}^{pre} \alpha s_j s_k, & j \neq k. \end{cases} \quad (328)$$

After a merger, some off-diagonal elements that were zero become negative. This increases the internalized diversion between the merging products and usually raises the fixed point of the pricing equations.

A.11 Chapter 5 Addendum: Why Demand Rotation Identifies Conduct

A demand shifter moves the demand curve outward or inward. A demand rotator changes the slope. In the conduct FOC

$$P(Q) + \lambda Q P'(Q) = MC(Q), \quad (329)$$

only $P'(Q)$ multiplies λ . If all observed variation only shifts $P(Q)$ while keeping $P'(Q)$ fixed at each Q , a change in price can be reinterpreted as a cost change. If an exogenous variable changes $P'(Q)$ at a fixed Q , then a firm with market power changes price because marginal revenue changes. A price taker with $\lambda = 0$ does not respond to the rotation through the markup term. This is the source of conduct identification.

A.12 Chapter 6 Addendum: Ordered Entry Probabilities

The homogeneous oligopoly model implies cumulative probabilities

$$\mathbb{P}(Y \geq y \mid x) = \Psi(v_y(x)). \quad (330)$$

The probability mass for exactly y firms is the difference between two adjacent cumulative probabilities:

$$\mathbb{P}(Y = y \mid x) = \mathbb{P}(Y \geq y \mid x) - \mathbb{P}(Y \geq y + 1 \mid x) = \Psi(v_y(x)) - \Psi(v_{y+1}(x)). \quad (331)$$

For the likelihood to be nonnegative, the model must imply

$$v_y(x) \geq v_{y+1}(x), \quad (332)$$

meaning per-firm variable profit weakly falls as the number of competitors increases.

A.13 Chapter 6 Addendum: The Multiple-Equilibrium Rectangle

In the two-firm complete-information game with $v_{1i}, v_{1j} < 0$, both asymmetric outcomes can be equilibria when

$$v_{0i}(x_i) + v_{1i}(x_i) \leq f_i \leq v_{0i}(x_i), \quad (333)$$

and

$$v_{0j}(x_j) + v_{1j}(x_j) \leq f_j \leq v_{0j}(x_j). \quad (334)$$

In this region, firm i wants to enter if firm j stays out, but wants to stay out if firm j enters; similarly for firm j . Therefore $(1, 0)$ and $(0, 1)$ are both self-confirming. Without a selection rule, the model predicts a set of possible outcomes rather than a single probability for each outcome.

A.14 Chapter 6 Addendum: Incomplete-Information Equilibrium as Fixed Point in Entry Probabilities

Let P_i be firm i 's equilibrium entry probability and suppose f_i has CDF F_i . From the threshold

$$t_i = v_{0i}(x_i) + v_{1i}(x_i)P_j, \quad (335)$$

we get

$$P_i = F_i(v_{0i}(x_i) + v_{1i}(x_i)P_j). \quad (336)$$

Similarly,

$$P_j = F_j(v_{0j}(x_j) + v_{1j}(x_j)P_i). \quad (337)$$

The equilibrium in beliefs is a fixed point in (P_i, P_j) . Multiple fixed points can arise when competitive effects are strong.

A.15 Chapter 7 Addendum: Bellman Contraction Proof Sketch

Let V and W be two value functions. For the logit Bellman operator,

$$(\Phi V)(s) = \log \sum_a \exp \left(\pi(a, s) + \beta \sum_{s'} g(a, s, s') V(s') \right) + \gamma. \quad (338)$$

The log-sum-exp function is Lipschitz with constant one in the sup norm:

$$\left| \log \sum_a e^{A_a} - \log \sum_a e^{B_a} \right| \leq \max_a |A_a - B_a|. \quad (339)$$

For each action,

$$\left| \beta \sum_{s'} g(a, s, s') [V(s') - W(s')] \right| \leq \beta \sum_{s'} g(a, s, s') \|V - W\|_\infty = \beta \|V - W\|_\infty. \quad (340)$$

Therefore

$$\|\Phi V - \Phi W\|_\infty \leq \beta \|V - W\|_\infty. \quad (341)$$

Since $\beta < 1$, the operator is a contraction.

A.16 Chapter 7 Addendum: Hotz-Miller Inversion for Logit

Under logit,

$$p(a | s) = \frac{\exp(v(a, s))}{\sum_k \exp(v(k, s))}. \quad (342)$$

Taking log odds relative to a reference action 0 gives

$$\log p(a | s) - \log p(0 | s) = v(a, s) - v(0, s). \quad (343)$$

Thus observed CCPs identify differences in choice-specific value functions. Because

$$v(a, s) - v(0, s) = \pi(a, s) - \pi(0, s) + \beta \sum_{s'} [g(a, s, s') - g(0, s, s')] V(s'), \quad (344)$$

normalizing $\pi(0, s) = 0$ turns CCP log odds into equations for payoff differences once the continuation value is expressed as a function of CCPs.

A.17 Chapter 7 Addendum: Partial Likelihood in NFXP

The full likelihood is

$$L(\theta) = \prod_t p_{\theta_1, \theta_2}(a_t | s_t) g_{\theta_2}(s_{t+1} | s_t, a_t). \quad (345)$$

Parameters θ_2 governing transitions can be estimated from

$$\hat{\theta}_2 = \arg \max_{\theta_2} \sum_t \log g_{\theta_2}(s_{t+1} | s_t, a_t), \quad (346)$$

without solving the dynamic program. Then estimate θ_1 from

$$\hat{\theta}_1 = \arg \max_{\theta_1} \sum_t \log p_{\theta_1, \hat{\theta}_2}(a_t | s_t). \quad (347)$$

This is less efficient than full maximum likelihood but computationally simpler.

A.18 Chapter 8 Addendum: From Individual CCPs to Joint Action Probabilities

If private shocks are independent across players conditional on the public state, each player's mixed action is independent conditional on s . Therefore

$$p(a | s) = \prod_{i=1}^N p_i(a_i | s). \quad (348)$$

This product structure is crucial for estimating joint action probabilities when the number of action profiles is large. It also means that a player's belief about rivals is summarized by

$$p_{-i}(a_{-i} | s) = \prod_{m \neq i} p_m(a_m | s). \quad (349)$$

The choice-specific value becomes

$$v_i(a_i, s) = \sum_{a_{-i}} p_{-i}(a_{-i} | s) \left[\pi_i(a_i, a_{-i}, s) + \beta \sum_{s'} g(a_i, a_{-i}, s, s') V_i(s') \right]. \quad (350)$$

A.19 Chapter 8 Addendum: Why Multiple Equilibria Complicate Likelihood

A likelihood requires the probability of the observed action conditional on the state and parameters. In a single-agent model, the optimal policy is unique under standard logit assumptions, so $p_\theta(a | s)$ is well-defined. In a dynamic game, the same θ may imply multiple Markov perfect equilibria $p^{(1)}, p^{(2)}, \dots$. The likelihood contribution would be

$$\mathbb{P}(a_t | s_t; \theta) = \sum_m \mathbb{P}(a_t | s_t, p^{(m)}, \theta) \mathbb{P}(p^{(m)} \text{ selected} | s_t, \theta), \quad (351)$$

but the equilibrium selection probabilities are usually not specified. The CCP approach conditions on the equilibrium observed in the data and estimates payoff parameters that rationalize it.

A.20 Chapter 8 Addendum: Forward Simulation for BBL Values

For a candidate policy profile σ and parameter θ , the value at state $s_0 = s$ is

$$V_i^\sigma(s) = \mathbb{E}_\sigma \left[\sum_{t=0}^{\infty} \beta^t \pi_i(a_t, s_t, \epsilon_{it}; \theta) | s_0 = s \right]. \quad (352)$$

Approximate it by R simulated paths of length T :

$$\hat{V}_i^\sigma(s) = \frac{1}{R} \sum_{r=1}^R \sum_{t=0}^T \beta^t \pi_i(a_t^{(r)}, s_t^{(r)}, \epsilon_{it}^{(r)}; \theta). \quad (353)$$

The truncation error is bounded if payoffs are bounded. If $|\pi_i| \leq \bar{\pi}$, then the remaining tail is at most

$$\sum_{t=T+1}^{\infty} \beta^t \bar{\pi} = \frac{\beta^{T+1} \bar{\pi}}{1 - \beta}. \quad (354)$$

A.21 Chapter 9 Addendum: Why First-Price Bids Are Below Values

The first-price FOC can be written as

$$x - b = \frac{H(b)}{h(b)}. \quad (355)$$

The right side is positive, so $b < x$. The bidder trades off margin against probability of winning. Raising the bid by a small amount lowers the surplus if the bidder wins, but increases the probability of winning. At the optimum, the marginal probability gain times surplus equals the marginal payment cost.

A.22 Chapter 9 Addendum: Observed Heterogeneity in Auctions

Suppose values are

$$x_{it} = z'_{it}\beta + u_{it}. \quad (356)$$

In a second-price IPV auction, bids reveal values, so

$$b_{it} = z'_{it}\beta + u_{it}. \quad (357)$$

One can estimate β and then study the residual distribution of u_{it} . In a first-price auction, bids are shaded:

$$b_{it} = \beta_t(x_{it}; z_{it}, n_t, r_t), \quad (358)$$

so one must either solve the equilibrium with covariates or use a homogenization step to condition on comparable auctions before applying the GPV inversion.

A.23 Chapter 9 Addendum: Winner's Curse in Common-Value Auctions

In a common-value environment, winning is bad news about rivals' signals: if bidder i wins, others likely had lower signals. A bidder with signal x should not bid the unconditional expected value $\mathbb{E}[V \mid X_i = x]$. In a symmetric second-price common-value setting, the dropout rule uses

$$b(x) = \mathbb{E}[V \mid X_i = x, Y_1 = x], \quad (359)$$

not $\mathbb{E}[V \mid X_i = x]$. Conditioning on $Y_1 = x$ accounts for the information revealed by just winning against the strongest rival.

B Compact Checklist of Estimation Moments

- Production IV: $\mathbb{E}[Z(y - X\beta)] = 0$.
- OP first stage: $\mathbb{E}[y - \beta_l l - \phi(k, i) \mid l, k, i] = 0$.
- OP second stage: $\mathbb{E}[v + \eta \mid k_t, k_{t-1}, i_{t-1}] = 0$.
- LP second stage: avoid conditioning on current materials; use predetermined variables.
- Logit demand with product fixed effects: $\xi_{jt} = \log s_{jt} - \log s_{0t} - x'_{jt}\beta + \alpha p_{jt}$ and $\mathbb{E}[Z_{jt}\xi_{jt}] = 0$.
- BLP: solve $s_t = \sigma_t(\delta_t, \theta_2)$, then impose $\mathbb{E}[Z_{jt}\xi_{jt}(\theta)] = 0$.
- Merger simulation: $mc = p - [\Omega^{pre}(p)]^{-1}s(p)$, then solve $p^{post} = mc + [\Omega^{post}(p^{post})]^{-1}s(p^{post})$.
- Dynamic single-agent CCP: $p = \Psi_\theta(p)$.
- Dynamic game CCP: $p = \Psi_\theta(p)$ with beliefs over rivals' actions.

- First-price auction GPV: $x = b + H(b)/h(b)$.

C Common Normalizations

- Static logit: normalize outside-good utility to zero.
- Random utility models: scale normalization is needed because utility is ordinal.
- Dynamic discrete choice: fix discount factor or provide exclusion restrictions; normalize one action payoff, often $\pi(0, s) = 0$.
- Entry models: normalize either the fixed-cost distribution or the scale of variable profit.
- Auctions: in IPV second-price auctions, bids reveal values; in first-price auctions, values require equilibrium inversion.